Summary

The recovery of rare earth elements (REEs) from magnet manufacturing waste is dominated by established chemical hydrometallurgical processes, with solvent extraction being the primary commercial method for achieving high-purity separated oxides. These incumbent technologies demonstrate high recovery rates, often exceeding 90-95%, and produce REE oxides with purities greater than 99.5%. However, they are often energy-intensive and rely on large volumes of organic solvents and acids. In contrast, biological recovery methods, utilizing microalgae, bacteria, and fungi, are emerging as a promising, environmentally-friendlier alternative, operating at ambient conditions with potentially lower carbon footprints.

Biological technologies are largely in the laboratory to pilot-scale development phase. Research on microalgae like *Chlorella vulgaris* shows significant biosorption capacities for neodymium (Nd), reaching up to 188 mg/g in lab studies using real magnet leachate. Engineered bacteria, particularly *E. coli* displaying lanthanide-binding proteins (Lanmodulin), demonstrate exceptional selectivity for REEs over other metals in magnet waste streams, with high regeneration potential over multiple cycles. Despite these advantages, biological methods face challenges in scaling up, maintaining performance in complex industrial waste streams, and competing with the high throughput and purity levels of established chemical processes.

The commercial landscape is active, with numerous companies operating or constructing chemical-based magnet recycling facilities across North America and Europe. Key players like HyProMag, Heraeus Remloy, and Cyclic Materials are deploying technologies such as hydrogen processing and advanced hydrometallurgy to produce recycled magnets and REE oxides at capacities of hundreds to thousands of tons per year. While several biotech startups are focused on biomining from ores and other waste, commercial-scale biological recovery of REEs specifically from magnet manufacturing waste is not yet established.

Key Findings

- **Chemical Technologies Dominate Commercially**: Incumbent hydrometallurgical processes, particularly solvent extraction using extractants like D2EHPA, are the only methods currently deployed at industrial scale for high-purity REE separation from magnet waste, achieving >99.5% pure products.
- **Biological Methods Show High Lab-Scale Potential**: Microalgae and bacteria demonstrate significant potential in controlled settings. *Chlorella vulgaris* has shown a biosorption capacity of over 150 mg/g for neodymium from real magnet leachate. Engineered *E. coli* with lanmodulin-binding proteins achieves high selectivity and can be regenerated for multiple cycles, retaining 85% of its capacity after nine cycles.
- **Scale-Up is the Primary Barrier for Biological Technologies**: While lab and bench-scale results are promising, biological methods are almost exclusively at a low Technology Readiness Level (TRL 3-4). Challenges include maintaining performance in complex industrial matrices, biomass production and harvesting logistics, and competing with the throughput of chemical plants.
- **Hydrogen Processing is an Emerging Commercial Alternative**: Technologies like HyProMag's Hydrogen Processing of Magnet Scrap (HPMS) offer a "short-loop" recycling route, breaking down magnets into a demagnetized powder for direct reuse in new magnets. This process claims an 88% reduction in energy use compared to primary production and is being deployed at commercial scale in the UK and Germany.
- **A Robust Commercial Recycling Market is Expanding**: Numerous companies, including HyProMag, Heraeus Remloy, Cyclic Materials, and Noveon Magnetics, have recently launched or are scaling up commercial magnet recycling facilities in Europe and North America, with capacities ranging from 100 to over 1,000 tons per year, driven by government incentives and OEM partnerships.
- **Selectivity and Regeneration are Key Differentiators**: Engineered biological systems, such as those using lanmodulin, show remarkable selectivity for REEs over transition metals like iron and cobalt found in magnet

waste. Chemical methods achieve selectivity through multi-stage solvent extraction or precipitation steps, which can be reagent-intensive.

- **Real Waste Stream Data is Limited for Biological Methods:** Most biological recovery studies use synthetic solutions. While some research has used real magnet leachates, data on performance with industrial grinding coolants or electroplating effluents, which contain interfering oils and surfactants, is scarce.

Methods

This technology landscape report was compiled through a systematic review of academic literature, patent databases, and web-based open-source intelligence. Research focused on identifying performance data for both biological and chemical technologies for recovering rare earth elements (REEs) from magnet manufacturing waste streams.

Academic sources were prioritized to gather empirical data on technology performance, including biosorption capacities, recovery efficiencies, separation factors, and kinetic parameters. Web searches were conducted to identify companies active in the magnet recycling and biological metal recovery sectors. Company websites, press releases, and industry news articles were analyzed to build profiles detailing each company's technology, commercial status, capacity, partnerships, and recent milestones between 2022 and 2026. This multi-pronged approach allowed for the synthesis of scientific data with current market and competitive intelligence.

Biological Technologies

Biological technologies for REE recovery leverage the natural ability of microorganisms like algae, bacteria, and fungi to bind or leach metals. These methods, primarily biosorption and bioleaching, are researched as a more sustainable alternative to conventional hydrometallurgy, often operating at ambient temperatures and neutral pH with minimal hazardous chemical use. While still in early stages of development for magnet waste applications, these technologies show significant promise in laboratory settings.

Microalgae

Microalgae are a key focus for biosorption due to their fast growth and the presence of functional groups on their cell walls that can bind metal ions. Various species have been investigated for their ability to adsorb REEs like neodymium (Nd) and dysprosium (Dy).

- ***Chlorella vulgaris*** has been extensively studied and demonstrates high biosorption capacities for Nd. Research using real leachate from Nd-Fe-B magnets reported a maximum experimental uptake of 157.40 mg/g at 35℃ and pH 5, with a theoretical maximum capacity (qmax) of 188.68 mg/g. The process is rapid, reaching equilibrium within 30 minutes, and follows pseudo-second-order kinetics. However, its performance is sensitive to temperature and biosorbent dosage.
- ***Spirulina platensis*** has been tested for Nd recovery from phosphogypsum leachate, showing a maximum capacity of 72.5 mg/g and reaching equilibrium in approximately 80 minutes.
- ***Galdieria sulphuraria***, a polyextremophile, has been assessed for REE biosorption, but studies show it has poor selectivity for REEs over other metals like copper in synthetic solutions. Its performance is also highly dependent on pH, with sorption improving significantly at pH 5 compared to more acidic conditions.

The table below summarizes the biosorption performance of several microalgae species for REE recovery.

Organism	Target REE	Max. Capacity (mg/g)	Optimal pH	Equilibrium Time	Waste Stream Tested
Chlorella vulgaris	Nd	157.4 (exp.), 188.7 (calc.)	5	30 min	Real Magnet Leachate
Spirulina platensis (LEB-18)	Nd	72.5	6	80 min	Phosphogypsum Leachate
Nostoc sp.	Ce, Nd, Tb, La	84.2 - 91.5	5	5 min	Synthetic Solution
Galdieria sulphuraria	La, Nd, Dy	2.4	5	>3 hours	Synthetic Solution

A key limitation for microalgae is the lack of robust desorption and regeneration data. Many studies do not report on the number of cycles a biosorbent can withstand before losing capacity, and some suggest combustion of the biomass to recover the metal, which prevents reuse. Furthermore, performance in complex industrial matrices containing oils, surfactants, and complexing agents remains largely untested.

Bacteria

Bacteria are being explored for both biosorption and bioleaching of REEs. Genetically engineered bacteria, in particular, offer a pathway to highly selective and efficient recovery systems.

- **Engineered *Escherichia coli***: A strain of *E. coli* engineered to display lanthanide-binding tags (LBTs) based on the protein lanmodulin on its cell surface has been successfully tested on real NdFeB magnet leachate. When encapsulated in hydrogel beads, this biosorbent showed a capacity of ~2.6 mg Nd/g and high selectivity, with REEs comprising 97% of the adsorbed metals at breakthrough. The system is highly regenerable, retaining 85% of its capacity after nine adsorption-desorption cycles using a mild citrate solution as the eluent.
- **Bioleaching Microbes**: Acidophilic and other bacteria can produce organic acids that leach REEs from solid magnet scrap. While studied for various e-wastes, specific data on bioleaching efficiency for Nd/Dy from magnet manufacturing waste is less common in the reviewed literature. Companies like REEgen are commercializing genetically optimized bacteria that produce biodegradable acids for extracting REEs from various feedstocks, including e-waste and industrial slags.
- ***Serratia sp.***: Biofilms of *Serratia sp.* are being examined for their potential to recover neodymium, with a focus on their tolerance to low pH and high salt concentrations found in leachates.

The primary advantage of engineered bacterial systems is their potential for high selectivity, which is a major challenge for many other biosorbents. However, like microalgae, the current demonstrated scale is limited to bench-top reactors, and the cost of producing and immobilizing engineered organisms needs to be considered for industrial applications.

Fungi

Fungal bioleaching is another area of interest, primarily using species like *Aspergillus* and *Penicillium*. These fungi produce organic acids, such as citric and oxalic acid, which can effectively leach Nd and Dy from magnet powders. This approach is positioned as a greener alternative to harsh inorganic acids like sulfuric acid. However, quantitative comparisons of fungal bioleaching versus chemical leaching for magnet waste are not well-documented in the available sources.

Chemical Benchmarks

Chemical technologies represent the incumbent and commercially established methods for REE recovery from various sources, including magnet waste. These processes are generally characterized by high recovery rates and the ability to produce high-purity products, though often at the cost of high energy consumption and significant chemical waste generation.

Solvent Extraction

Solvent extraction (SX) is the industry-standard technology for separating individual REEs to high purity (up to 99.99%) and is the only method used on a large commercial scale for this purpose.

- **Extractants**: The most common extractants are organophosphorus acids like Di-(2-ethylhexyl) phosphoric acid (D2EHPA), which is versatile and commercially available. Other extractants such as Cyanex 272 and ionic liquids are also used to achieve specific separation goals.
- **Performance**: D2EHPA can achieve over 99% extraction of REEs from real magnet leachates, with stripping efficiencies also approaching 100% using mineral acids like HCl. Separation factors between adjacent REEs are typically low (around 2.5), necessitating many extraction stages for high-purity separation. However, systems can be optimized for specific pairs; for example, separation factors for Dy/Nd can reach over 200 with D2EHPA under specific conditions.
- **Limitations**: The primary drawbacks of SX include the large volumes of organic solvents and acids required, significant wastewater generation, and the high number of stages needed, which increases capital and operational costs.

Acid Leaching and Precipitation

This is a fundamental hydrometallurgical route where magnet scrap or sludge is first dissolved in strong mineral acids, followed by selective precipitation of REEs.

- **Leaching**: Acids like sulfuric acid (H2SO4), hydrochloric acid (HCl), and nitric acid (HNO3) are used to dissolve magnet waste, with leaching efficiencies for Nd and Dy often exceeding 90-99%. Pre-treatment steps like roasting can improve the selectivity of REE leaching over iron.
- **Precipitation**: REEs are commonly precipitated from the leach solution as oxalates or hydroxides to separate them from other dissolved metals. This step can yield a mixed REE concentrate with high purity; for instance, a process involving selective leaching and precipitation from NdFeB sludge recovered 98.28% of the REEs, and another process produced a mixed REE oxide with 98.4% purity.

Other Chemical Technologies

- **Ion Exchange**: Resin-based separation is used for REE extraction, but achieving high purity separation between adjacent REEs is difficult due to their similar chemical properties. Companies like Multotec offer continuous ion exchange systems (Clean-iX®) claiming >90% REE recovery.
- **Electrochemical Methods**: Electrodeposition and electrowinning are being explored. Rare Earth Salts has developed an electrowinning process, and the EU-funded REE4EU project demonstrated high-temperature electrolysis for producing REE alloys from waste.
- **Membrane Technologies**: Nanofiltration and other membrane-based systems are primarily used for concentrating REEs in leach solutions before further processing, rather than for high-purity separation.

Company Landscape

The commercial landscape for REE recovery from magnet waste is dynamic, with established hydrometallurgical companies, specialized recycling firms, and emerging biotech startups all vying for position. While chemical-based recycling is already at a commercial scale, biological methods remain largely in the pilot and venture-funded startup phase.

Magnet Recycling Companies (Chemical/Physical)

Several companies are operating or commissioning commercial-scale facilities using non-biological methods, primarily advanced hydrometallurgy and hydrogen-based processing.

Company	Headquarters	Core Technology	Target Waste Streams	Claimed Capacity	Commercial Status
HyProMag	UK, Germany, USA	Hydrogen Processing of Magnet Scrap (HPMS)	End-of-life magnets from HDDs, EVs, wind turbines	100-300 tonnes/year (UK plant)	Commercial plant operational in UK (2026); plants under development in Germany and USA
Cyclic Materials	Toronto, Canada	MagCycle℠ & REEPure℠ (Hydrometallurgy)	EV motors, wind turbines, HDDs, manufacturing scrap	600 tonnes/year (planned, expandable)	Pilot/demo facilities operational; investing in commercial plants in the US and Canada
Heraeus Remloy	Hanau/Bitterfeld, Germany	Chemical-free recycling into nanocrystalline magnetic powders	E-waste, computer hard disks	600-1,200 tons/year	Commercial plant operational in Bitterfeld, Germany (Europe's largest)
Noveon Magnetics	San Marcos, Texas, USA	Magnet-to-magnet recycling via powder metallurgy (EcoFlux™)	HDDs, EV motors, MRI machines, scooters	~2,000 tons/year (target)	Commercial facility operational in Texas
Geomega Resources	Boucherville, QC, Canada	Closed-loop hydrometallurgy	Magnet scrap and swarf	1.5 tonnes/day (demo plant)	Demonstration plant (TRL 7-8) under construction

Company	Headquarters	Core Technology	Target Waste Streams	Claimed Capacity	Commercial Status
REEcycle	Houston, TX, USA	Proprietary solvent extraction (patented)	NdFeB magnets from e-waste, EVs, wind turbines, swarf	99.8% recovery efficiency claimed	Commercial-facing operations

Biological Metal Recovery Companies

While many companies focus on biomining from primary ores, a few startups are specifically targeting REE recovery from waste streams, positioning biotechnology as a future solution.

- **REEgen, Inc.**: An Ithaca, NY-based startup commercializing technology from Cornell University and Idaho National Laboratory. It uses genetically-engineered bacteria for bioleaching REEs from industrial wastes like coal ash, steel slag, and e-waste, claiming an efficient, low-cost process without harsh chemicals. The company has received NSF SBIR funding and won commercialization competitions in 2025 to accelerate its path to market.
- **BiotaTec**: This company is developing and licensing biomining technologies for extracting critical metals from low-grade ores, tailings, and WEEE. It has received funding from the European Innovation Council to advance its BiotaMet platform.
- **Alta Resource Technologies**: Backed by venture capital and government entities like DARPA and the DOE, Alta is developing a platform using engineered proteins, based on lanmodulin, for highly selective separation of REEs from e-waste, mining tailings, and ores. The technology, licensed from Lawrence Livermore National Laboratory, claims the ability to produce >99% pure products.
- **BIOWEG**: In partnership with TU Berlin, this German startup secured \u20ac1.5 million from SPRIND in 2025 to develop a bio-leaching platform using bioacids from fermentation and peptide-based separation for REE recovery from water-based streams.

Traditional Hydrometallurgy and Technology Providers

Established mining and chemical companies, along with specialized equipment providers, form the backbone of the existing and expanding chemical-based REE supply chain.

- **Lynas Rare Earths & MP Materials**: These are major global producers of separated REEs from primary mining operations. Both are expanding their processing capabilities in the U.S. with government support, with plans that include recycling feedstocks in addition to primary concentrates.
- **Umicore**: A major materials technology and recycling group, Umicore has partnered with automakers like BMW to establish dedicated rare earth recycling facilities, focusing on a circular economy for EV motor magnets.
- **Technology & Equipment Providers**: Companies like **Koch Modular** and **Tenova** provide commercial-scale solvent extraction plants and modular systems. The **Saskatchewan Research Council (SRC)** has developed a commercial-scale REE processing facility and offers turnkey services, including proprietary solvent extraction cell technology.

The landscape shows a clear trend towards establishing regional, secure supply chains for REEs, with significant government funding and strategic partnerships between recyclers, technology developers, and end-users like automotive OEMs and electronics manufacturers.

Disclaimer: This report is based on publicly available information and data extracted from the provided sources. Performance metrics and company statuses are as reported in those sources and may have evolved.

Customer landscape and market-entry risks for biological REE recovery in permanent magnet manufacturing

Summary

The permanent magnet market is large and growing, with one market estimate projecting growth from USD 21.98B (2025) to USD 30.01B (2030). Another industry snapshot values the rare-earth magnet market at ~US$19.5B in 2024, indicating substantial addressable value for technologies that reduce cost, improve resilience, or support recycling claims.

Supply and manufacturing remain highly concentrated in China, with multiple sources estimating ~85–90% (or ~92%) of global magnet manufacturing/production occurring in China, creating acute geopolitical and supply-chain risk that is already disrupting downstream OEMs and Tier-1s. Demand pressure is driven by EVs, renewables, robotics, and other electrification end markets, reinforcing the strategic importance of NdFeB and, for defense, SmCo magnets.

For biological REE recovery solutions aimed at magnet manufacturers, the most consistently evidenced customer value drivers in the provided sources are (1) high and rising waste volumes and material losses in NdFeB production, (2) mounting compliance and waste-treatment costs, and (3) supply-security premiums being paid for non-China magnets and recycled content—especially by defense and automotive ecosystems. The most consistently evidenced barriers and risks are (1) scale-up risk between lab and pilot, (2) matrix complexity and competing ions that degrade selectivity and increase biomass/pretreatment needs, and (3) commercialization viability versus low-priced Chinese supply and price volatility in rare earths.

Customer segmentation

The customer landscape for biological REE recovery spans (a) NdFeB magnet manufacturers (sintered and bonded), (b) SmCo magnet manufacturers (niche, defense-relevant), and (c) vertically integrated OEMs and Tier-1 suppliers that consume magnets and may influence specifications and supplier qualification.

NdFeB magnet manufacturers

A core segmentation insight is the sheer scale of NdFeB production and the magnitude of internal manufacturing losses: global NdFeB production is estimated at ~130,000–150,000 t/y, and NdFeB manufacturing waste is estimated at ~30 wt% of total production, implying large, consistent "new scrap" feedstocks in the form of sintering fragments, cutting scraps, grinding powders and sludge. NdFeB magnets also contain approximately 30–32% REEs, underscoring the embedded material value in these waste streams and the potential for recovery-driven economics if separation and polishing can meet magnet-grade specifications.

The table below consolidates magnet manufacturers and projects explicitly named in the provided sources, emphasizing locations and any stated capacities.

Company or project	Segment	Geography	Capacity signal	Notes relevant to REE recovery targeting
Ningbo Tongchuang Magnetic Materials	NdFeB manufacturer	China (Ningbo)	Expansion to 10,000 t/y annual production capacity.	Capacity disclosure suggests sizable, centralized waste generation potential, but facility waste volumes are not reported in the source.
ZHAOBAO MAGNET	NdFeB manufacturer	China	Annual production capacity 8,000 t/y.	Capacity disclosure suggests sizable "new scrap" streams, but the source does not quantify wastewater or residue volumes.
Neo Performance Materials (Estonia plant)	NdFeB (sintered) manufacturer	EU (Estonia)	Initial capacity 2,000 t/y expanding to ≥5,000 t/y.	Neo explicitly targets recycling of end-of-life magnets and "swarf" from manufacturing, indicating internal appetite for closed-loop recovery solutions.
MP Materials (Fort Worth)	NdFeB manufacturer	US (Texas)	First phase 1,000 t/y capacity.	Part of reshoring/industrial policy response to supply risk, with potential preference for domestic circular feedstocks.
Noveon Magnetics	NdFeB manufacturer	US (Texas)	~2,000 t/y capacity.	U.S.-based capacity could be a near-term pilot beachhead for onsite recovery given heightened supply-chain concerns.

Company or project	Segment	Geography	Capacity signal	Notes relevant to REE recovery targeting
USA Rare Earth	NdFeB manufacturer	US (Oklahoma)	~1,200 t/y initially, ramping to 4,800 t/y by ~2026.	Scale-up plans create commissioning windows where new waste-treatment systems may be designed-in rather than retrofitted.
GKN Powder Metallurgy (planned)	NdFeB capacity project	EU/NA (planned)	Planned 4,000 t/y magnet capacity.	New-build projects can potentially adopt novel recovery processes if permitting and risk-sharing are addressed early.
JL MAG, Ningbo Yunsheng, San Huan (Zhongke Sanhuan), Tianhe, ZHmag, Earth-Panda	NdFeB manufacturers (major Chinese cohort)	China	Identified as dominant global suppliers; capacities described as "tens of thousands of tons" for top firms in China-centric sources.	China's ecosystem advantage and scale create strong incumbent competition; market entry may require differentiated compliance/ESG or supply-security value.

Beyond these examples, multiple market listings identify additional key magnet-industry players (e.g., Proterial, Shin-Etsu, TDK, VACUUMSCHMELZE, Zhongke Sanhuan, Yantai Zhenghai), but the provided excerpts do not include capacity-by-site data for most of them.

SmCo magnet manufacturers

SmCo magnets are consistently positioned as strategically important for defense applications, even if they represent a smaller share of magnet volume in the broader market segmentation. Export-control discourse also flags samarium as a controlled element due to its relevance for SmCo, implying feedstock risk and higher willingness to pay for secure, compliant supply chains in this niche.

Because the provided sources do not enumerate major SmCo producers with capacities, a practical segmentation approach (supported by the sources) is to treat SmCo as a "defense-critical, supply-sensitive" submarket where domestic-content rules and export restrictions can outweigh pure cost minimization.

Vertically integrated OEMs and Tier-1s

Automotive OEMs and Tier-1 suppliers are key "specification setters" and indirect customers because they ultimately drive magnet qualification, supply-security requirements, and recycled-content or traceability expectations in the value chain. The automotive ecosystem evidence in the sources explicitly names OEMs (Ford, GM, Toyota, Volkswagen) and Tier-1s (Bosch, ZF, Denso, BorgWarner, Marelli, Schaeffler) as the integrators of magnets into motors and electronics, implying they influence acceptance criteria for recycled or recovered REEs.

EV motors are a major NdFeB demand sink, with a cited estimate that each EV typically contains ~3–5 kg of NdFeB magnets, connecting end-market growth directly to magnet production scale and associated in-process wastes. Wind energy is also a magnet-intensive end market, with one cited example indicating turbines can house "between one to eight tons" of permanent magnets depending on size, which underpins long-term end-of-life recycling demand and circular procurement programs.

Waste management pain points

Waste management pain points for magnet manufacturers and adjacent REE-processing operations cluster into (1) waste and effluent burdens intrinsic to current recycling and hydrometallurgy pathways, (2) compliance and permitting obligations, and (3) the cost and liability of secondary wastes such as sludge, along with broader ESG and supply-chain pressures that elevate the value of circularity.

Waste generation and process burdens

NdFeB manufacturing waste is explicitly estimated at ~30 wt% of total production, and the identified waste categories include sintering fragments, cutting scraps, grinding powders, and sludge, all of which can plausibly appear as solids or solid-laden liquids depending on plant practices. Hydrometallurgical recycling routes are described as generating "large volumes of intractable wastewater containing heavy metals and residual acids," which directly frames a target niche for lower-chemical-intensity biological capture or polishing steps (if selectivity and robustness are proven).

Multiple sources highlight that hydrometallurgical processing generates substantial sludge and wastewater, and that the broader rare-earth supply chain can generate very large waste burdens (e.g., "2,000 tons of toxic waste" per ton REE as cited in one source), amplifying the perceived liability and ESG exposure associated with REE processing and recovery.

Regulatory compliance and permitting

China's rare earth industry discharge regime (GB 26451-2011) explicitly defines major wastewater pollutants as suspended solids, COD, ammonium nitrogen, fluorides, heavy metals, and radioactive materials, and sets discharge limits by pollutant and process category, which indicates that compliance is multi-parameter and not solely driven by REE concentrations. General Chinese discharge standards also define numeric limits for common metals such as copper, zinc, and selenium (and a pH range 6–9), illustrating the typical compliance envelope that magnet operations with plating or metal-containing effluents must navigate even when Nd/Dy limits are not explicitly cited in the provided excerpts.

In the EU, the Industrial Emissions Directive framework emphasizes permits with emission limit values, monitoring, and operation consistent with best available techniques, which raises the compliance and documentation burden for any new onsite treatment technology that changes emission profiles. The same framework provides a mechanism for temporary derogations to test emerging techniques, implying that well-designed pilots with robust monitoring plans can be permitted, but only within a controlled and time-limited compliance pathway.

For the U.S., the provided excerpt from 40 CFR Part 421 Subpart Y specifies applicability to discharges from primary rare earth metals production facilities, indicating that wastewater standards can be codified at the subsector level, but the excerpt does not provide Nd/Dy numeric limits for magnet manufacturing specifically.

Compliance costs and liability

Sector-level evidence from China indicates that stricter environmental standards for rare earth production can drive very large compliance investments, with an estimate of 1.5 billion yuan for sewage treatment facilities plus annual running costs of 280 million yuan (and additional waste-gas treatment investments), indicating that compliance can be capital-intensive and structurally reshapes the industry through consolidation pressure. The same source notes historical ammonia nitrogen discharge concentrations in the hundreds to thousands of mg/L, "scores of times higher than the national standard," illustrating the magnitude of past compliance gaps and the likelihood that wastewater treatment upgrades remain a recurring imperative for REE-related industrial operations.

Across industrial wastewater contexts, sludge disposal is repeatedly characterized as a major hidden expense, with cited ranges of $150–600 per dry ton and potentially $800–2,000 per dry ton for hazardous classification, which is directly relevant to magnet waste treatment because many REE-bearing residues and metal sludges can be regulated as hazardous depending on jurisdiction and composition. The long-term liability of remediation and treatment can be extreme in worst cases, as illustrated by a site case study showing a settlement exceeding $950M and a treatment plant expected to operate "in perpetuity," underscoring why many operators prioritize solutions that minimize long-lived waste liabilities.

Sustainability and circularity pressures

Several sources frame recycling and recovery as a response to scarcity and geopolitical risk, not only as a cost play, such as an EU-linked wind operator initiative that emphasizes REE scarcity, sustainability impacts, and geopolitical supply risks as drivers for permanent magnet recycling programs aligned with EU critical raw materials policy. Corporate disclosures from a magnet producer (Neo) similarly emphasize investments in air, water and waste emissions control technologies and explicitly highlight recovery and recycling of magnet manufacturing "swarf" to support circular economics, indicating that at least some manufacturers already view internal scrap recovery as strategically important.

Willingness to pay and value

Willingness to pay for biological REE recovery is best evidenced indirectly in the provided sources through (a) what industrial actors pay for wastewater treatment capacity and compliance, (b) what downstream sectors pay for supply-security, and (c) baseline processing costs and compliance adders for REE recycling pathways that create headroom for lower-waste, lower-chemical alternatives.

Treatment cost baselines

Industrial wastewater treatment investments can be substantial, with one cited range of $2,000–$15,000 per m^3 of treatment capacity depending on contaminant complexity, implying that onsite solutions for metal-bearing effluents can reach multi-million-dollar capital programs for medium-to-large flows. The same source provides an example operating cost of ~$8.50 per m^3 treated, which is a useful benchmark when estimating the operating-cost headroom for a recovery system that adds pretreatment, nutrients/biomass handling, regeneration chemicals, and monitoring.

Other vendor-facing sources similarly cite that wastewater treatment projects can range from ~$400,000 to >$1.5M for complex multi-stage systems, and that stricter discharge limits typically require additional treatment steps or equipment, reinforcing the idea that compliance-driven "must spend" budgets exist in industrial settings even before adding recovery value.

Recycling and recovery cost baselines

For REE recycling via existing hydrometallurgical approaches, one source cites processing costs of $15–35 per kg and an additional $2–4 per kg attributable to acid disposal and wastewater treatment, making waste treatment and compliance a meaningful fraction of total cost. The same source also cites processing costs of $8–15 per kg REE recovered and feedstock acquisition costs of $0.50–2.00 per kg, illustrating the economic structure that a biological solution must beat or complement (e.g., by reducing waste-treatment costs, improving selectivity, or enabling onsite recovery that lowers logistics costs).

Supply-security premiums and long-term offtake

Downstream willingness to pay for non-China supply is explicitly evidenced by reports that buyers pay premiums of ~10–30% for non-Chinese magnets, and that some clients pay 15–20% more for Vietnamese magnets to reduce China exposure, indicating a real premium pool tied to supply resilience rather than purely to magnet cost. Defense demand is also described as willing to pay premium prices for non-Chinese sources, and U.S. government funding to support domestic magnet supply chains is cited at >$439M since 2020, suggesting that public-sector de-risking can support higher-cost domestic and circular solutions in strategic markets.

OEM willingness to underwrite recycled-content supply is also evidenced by a cited example of a $500M long-term agreement to source magnets manufactured in Texas from 100% recycled feedstock, which signals that large buyers may finance or guarantee demand for circular magnet supply when it aligns with strategic and ESG priorities.

Recovery value framing

A practical value framing for biological recovery from magnet manufacturing streams is to tie recovered REEs to market prices and specification requirements, while accounting for pretreatment and purification costs implied in the biosorption literature. One cited price point for neodymium oxide (~$77,983/metric ton as of Dec 2020) provides an anchor for estimating gross recovered value if concentration and recoverable mass are measured at a target facility. However, multiple sources stress that profitability depends strongly on REE concentration, feedstock composition, and pretreatment and waste-management costs, making facility-specific sampling and piloting a prerequisite for credible ROI claims.

Adoption barriers

Adoption barriers for biological REE recovery in this customer set are best evidenced as a combination of (1) proof and pilot requirements, (2) process integration constraints (space, monitoring, retrofit complexity), and (3) technical performance uncertainty in real multi-component wastes, plus a market barrier tied to economics versus cheap Chinese supply and technology substitution in end markets.

Proof requirements

Multiple sources describe the need for pilots and proof-of-concept before scaling or investment decisions, including an industrial sustainability disclosure stating that additional investments will be made only once "sufficient proof

of concept has been established." In the magnet supply chain context, pilots and testing are also described as part of supplier qualification, with examples of OEMs testing magnets under extreme conditions and renewable energy companies piloting magnets for durability, which is a strong analog for how a novel waste-treatment or recovery technology will likely need to prove reliability and consistency before deployment.

In the EU regulatory context, the Industrial Emissions Directive explicitly provides a pathway for testing emerging techniques via temporary derogations from BAT-associated emission levels, indicating that pilots are feasible but must be carefully scoped, monitored, and time-bounded within permitting regimes.

Integration and retrofit friction

Industrial wastewater sources emphasize hidden expenses and practical complexity associated with integrating and installing a new system into ongoing operations, reinforcing that footprint, tie-ins, and installation disruption can be major barriers for onsite biological systems unless modularized or designed into new builds. Cost-focused sources also cite that installation and construction may represent 30–45% of total project investment, and that installation/commissioning can be 15–40% of total cost in some contexts, making retrofits materially more expensive than "equipment-only" numbers suggest.

Permitting and compliance processes can also create material schedule risk, with one cited range of 6–18 months and $50,000–$500,000 for major projects, which directly affects adoption timelines and the attractiveness of phased pilots versus full-scale deployments.

Technical performance uncertainty in real wastes

Biological and biosorption approaches face well-documented performance risks in complex, multi-metal streams, including competition for binding sites in multi-metal solutions, reduced performance and higher required contact times, and sensitivity to operating conditions and matrix chemistry. Regeneration and desorption cycles can degrade capacity, with one cited example that acid washes can reduce adsorption capacity by ~50%, which directly impacts lifecycle economics and the number of economically viable reuse cycles for a biosorbent or biomass system.

For REE-specific leachates, one study highlights that non-REE concentrations (e.g., Na, K, Mg, Ca, Fe) can be orders of magnitude higher than REEs, and that Ca and Mg are primary factors affecting REE recovery efficiency and purity, implying that real magnet-related streams with hard-water ions, plating salts, or process additives could strongly influence performance unless pretreatment or selective binding is robust.

Some sources also emphasize that bio-hydrometallurgical approaches remain at early research stages (TRL 3–4) with 5–15 year commercialization timelines, which is a strong signal that magnet manufacturers will require compelling pilot evidence and risk-sharing before relying on biological systems for compliance-critical operations.

Market and organizational barriers

Commercial viability versus cheap Chinese imports is repeatedly cited as the primary obstacle for REE magnet recycling in the EU context, indicating that many customers will default to low-cost supply unless the solution also delivers compliance savings, premium pricing, or supply-security value that is monetizable through contracts. Industry sources also highlight that switching supply chains has historically been difficult due to delays, import fees, and customer support gaps, which is an analog barrier for any new third-party recovery service that requires operational trust and dependable logistics.

At the downstream level, technology substitution is a non-trivial barrier to the long-term market size for recovered REEs, because automakers are exploring magnet-free alternatives and designs that reduce rare-earth intensity, even

while permanent-magnet motors still dominate current EV production.

Strategic risks and mitigation

Strategic risks for market entry and scaling biological REE recovery can be organized into technology, market, operational, and strategic timing risks, with mitigations grounded in evidence from scale-up literature, wastewater implementation practice, and observed procurement behavior in analogous sectors.

Risk assessment

The table below provides an evidence-based risk register with likelihood and impact framed strictly from the provided sources.

Risk	Category	Likelihood	Impact	Mitigation and early indicators
Lab-to-pilot scale-up failure or major rework	Technology	High, because 60–80% of cleantech that reaches lab proof-of-concept encounters significant technical or economic challenges at pilot scale.	High, because pilot setbacks extend timelines and can force redesigns; bio-based approaches are often still TRL 3–4 with long commercialization horizons in some sources.	Use staged development and explicit scale-up criteria (as described for BioCOP) and gate capital release on measured performance at each scale. Early indicator: inability to maintain performance in continuous systems versus batch tests (noted as a common research gap).

Risk	Category	Likelihood	Impact	Mitigation and early indicators
Reduced selectivity and capacity in complex matrices (competing ions, non-REE metals)	Technology	High, because multi-metal competition reduces performance and increases required contact time, and non-REEs in leachates can be orders of magnitude higher than REEs.	Medium to high, because purity requirements for magnets are strict (>99.5%) and impurities such as Ca/Mg reduce recovery efficiency and purity, potentially requiring additional polishing steps.	Mitigate by measuring matrix composition and designing pretreatment and pH windows; profitability and performance are explicitly driven by feedstock composition and pretreatment costs. Early indicator: rising Ca/Mg levels correlating with product purity drift.
Biosorbent regeneration losses limiting lifecycle economics	Technology	Medium to high, because acid washes can reduce adsorption capacity by ~50% and reuse cycles may be limited depending on sorbent system.	Medium, because fewer cycles increase unit cost and waste generation from spent biomass/sorbent handling.	Design low-damage regeneration chemistry and track capacity decay per cycle; incorporate desorption to extend sorbent life where feasible (noted as an objective in continuous bioremoval systems). Early indicator: cycle-to-cycle capacity loss exceeding modeled thresholds (e.g., 10–20% per cycle).

Risk	Category	Likelihood	Impact	Mitigation and early indicators
Compliance and permitting delays for onsite systems	Operational	Medium, because permit and compliance expenses are cited at 6–18 months for major projects, and EU permitting emphasizes monitoring and BAT alignment.	Medium to high, because delays reduce time-to-revenue and may miss customer shutdown windows for retrofits, increasing installed cost.	Pilot under temporary derogations for emerging techniques where available and build monitoring plans into designs. Early indicator: extended data requests or unclear permit applicability for indirect discharges.
REE price declines reduce recovery attractiveness	Market	Medium, because depressed rare earth prices are cited as a challenge for economic viability of processors and profitability in China is described as being squeezed at low prices.	High, because multiple sources state recycling economics can be worse than recovered value, and price declines can make recycling "less attractive."	Structure contracts around compliance savings and supply-security value (not only recovered REE value), and target customers already paying 10–30% premiums for non-China supply. Early indicator: widening spread between processing cost ($15–35/kg) and REE product price benchmarks.

Risk	Category	Likelihood	Impact	Mitigation and early indicators
Competition from incumbents and cheap Chinese supply	Market	High, because recycled/local NdFeB magnets are cited as unable to compete on price with cheap Chinese imports, and Chinese-made NdFeB is expected to be 20–30% cheaper than nascent U.S.-made magnets in one source.	High, because price undercutting can delay customer adoption even if the technology works, making commercial viability the limiting factor for recycling scale-up in some analyses.	Target segments where premiums exist (defense; supply-secure OEM programs) and emphasize compliance cost avoidance and hazardous sludge reduction (high sludge disposal costs are cited). Early indicator: procurement decisions shifting back to lowest-cost imports after export controls ease.
Demand erosion from magnet-free motors or reduced-REE designs	Strategic	Medium, because automakers are exploring magnet-free alternatives and designs that reduce REE intensity, though permanent-magnet motors still dominate current EV production.	Medium, because reduced Nd/Dy intensity reduces recoverable mass per unit and shrinks addressable market over time, even if absolute magnet demand grows.	Focus on near-term "new scrap" feedstocks (manufacturing waste at ~30 wt%) that exist regardless of end-of-life flows, and diversify into adjacent dilute REE wastewater sources if needed (wastewater recovery is framed as economically attractive in some contexts). Early indicator: OEM procurement shifting to alternative motor architectures at scale.

Mitigation patterns from analogous deployments

Mining-sector bioleaching provides a concrete scale-up precedent: BioCOP progressed through staged development from 10 L pilot trials to a 200 t/day prototype plant, and the source emphasizes that staged development helps define scale-up criteria and mitigates commercialization risks. This staged pattern is directly transferrable to biological REE recovery for magnet wastes, implying that a "10–1000 L → skid pilot → full-scale module" trajectory with explicit go/no-go gates is likely to be required for conservative industrial customers.

Long-lived treatment liabilities also illustrate why customers may value solutions that reduce perpetual O&M burdens: a remediation case study describes a lime-neutralization/high-density sludge plant operating "in perpetuity," which implies that any technology that meaningfully reduces sludge volumes or enables source control can be attractive even if initial costs are higher.

Conclusions

The most defensible near-term customer beachhead for biological REE recovery, based on the provided evidence, is NdFeB manufacturing "new scrap" streams, because manufacturing waste is estimated at ~30 wt% of production and includes readily collectable categories such as grinding powders and sludge. A second defensible beachhead is supply-security-driven ecosystems (defense and automotive) that are already paying 10–30% premiums for non-China magnets and are supported by significant public funding to build domestic capacity, creating potential tolerance for higher-cost recovery solutions if they improve resilience or compliance outcomes.

However, the evidence also indicates that commercial viability versus low-cost Chinese supply is often the binding constraint, and that biological approaches face high technical uncertainty in scale-up and in complex real matrices, requiring pilots, staged development, and careful contract structures that monetize compliance savings and supply-security premiums rather than relying on REE commodity prices alone. Any market-entry plan should therefore be organized around (1) rapid sampling and matrix characterization (to validate concentration and interfering ions), (2) pilots designed to satisfy permitting and monitoring regimes, and (3) business models that share risk and link fees to measurable compliance and recovery outcomes, consistent with the recurring emphasis on proof-of-concept, monitoring, and lifecycle cost considerations in industrial wastewater decision-making.

REE Recovery Technology Landscape for Magnet Manufacturing Waste (2015–2026)

Summary

This landscape maps biological versus incumbent chemical approaches for recovering rare earth elements (REEs) from magnet-related wastes, emphasizing evidence that uses real NdFeB magnet leachates or end-of-life magnet feedstocks. The strongest magnet-relevant biological evidence in the provided sources comes from (1) high-capacity neodymium (Nd) biosorption using dried *Chlorella vulgaris* on NdFeB-derived leachates and (2) engineered *E. coli* displaying lanthanide-binding tags (LBTs), validated on sulfuric-acid leachates of end-of-life NdFeB magnets in batch and in a bench-scale packed bed column. *C. vulgaris* achieved maximum Nd uptake of 157.21 mg/g at pH 5 in a mixed leachate derived from neodymium magnets and also showed a Langmuir qmax of 188.68 mg/g on a solution derived from real Nd–Fe–B magnets (hard disk magnets), with pseudo-second-order kinetics and equilibrium within 30 minutes under the tested conditions. LBT-displayed *E. coli* extracted REEs from NdFeB magnet leachate with 91% extraction at 95% purity in small batch assays and showed strong REE selectivity in a 77 mL column, where REEs were 97% of adsorbed metals by mass at breakthrough; it also demonstrated ~94% Nd desorption with 10 mM citrate and retained 85% capacity after nine cycles.

Chemical benchmarks remain dominated by hydrometallurgical processing, with solvent extraction identified across multiple sources as the most useful/most successful industrial separation route for REEs from complex aqueous matrices, capable of producing very high purity outputs (up to 99.99% in cited sources). For magnet leachates, organophosphorus extractants such as D2EHPA show rapid equilibration on the order of minutes and can achieve complete transfer of REEs to the organic phase under some conditions, with quantitative examples including 100% transfer at 0.9–1.2 M D2EHPA in hexane/octane and 100% stripping at ≥2 M HCl in cited tests. Ionic liquids are presented as alternative extractant systems, including cases reporting final oxide purities of 99.6% Nd2O3 and 99.8% Dy2O3 and high Nd/Co and Dy/Co separation factors (αNd/Co 590; αDy/Co 500) in the cited NdFeB-related separation workflow.

Commercial deployment for magnet recycling in the provided sources is primarily via physical/direct recycling and hydrometallurgical circuits, with multiple companies claiming operational plants, demonstration facilities, or announced expansions. Examples include HyProMag's Hydrogen Processing of Magnet Scrap (HPMS) technology (commercial facility referenced in the UK and additional plants/targets referenced in Germany/USA), Cyclic Materials' MagCycle/REEPure platform with claimed recovery rates exceeding 98% and multiple facility announcements, and Heraeus Remloy's production start at a 600 t/y plant (expandable to 1,200 t/y) for nanocrystalline rare earth magnetic powders from recycled end-of-life devices. Biological commercialization is earlier-stage in the provided sources, but several companies are explicitly positioned as commercializing engineered-biology or protein-based separations (e.g., REEgen, Alta, BIOWEG).

Key Findings

The evidence base supports several decision-relevant conclusions about performance, readiness, and the biological–chemical trade space for magnet-related REE recovery.

- Dried *Chlorella vulgaris* biosorbed Nd from real magnet-derived leachates at high capacities, including 157.21 mg/g at pH 5 from a mixed leachate derived from neodymium magnets and a Langmuir qmax of 188.68 mg/g at pH 5 and 35 ℃ from a solution derived from real Nd–Fe–B magnets.
- In the hard-disk-magnet-derived Nd system, Nd sorption on *C. vulgaris* followed pseudo-second-order kinetics with equilibrium reached within 30 minutes and >90% of uptake occurring early in the time course under tested

conditions.

- A key realism gap for some magnet-leachate biosorption data is that one cited *C. vulgaris* hard-disk-magnet leachate preparation resulted in Fe, B, Dy, and Co below detection limits due to "selective leaching," limiting what can be concluded about Nd/Fe or Dy/Co selectivity under fully representative multi-metal magnet waste matrices.
- Engineered LBT-displayed *E. coli* demonstrated strong REE selectivity on real NdFeB leachate in column mode, with REEs comprising 97% of adsorbed metals by mass at breakthrough and minimal non-REE adsorption (transition metal breakthrough within one bed volume) in the cited column tests.
- Among biological approaches in the provided sources, regeneration durability is most explicitly quantified for the LBT-*E. coli* column platform, including ~94% Nd desorption using 10 mM citrate (pH 6) and 85% capacity retention after nine adsorption/desorption cycles.
- Microalgae/algae systems show strong pH dependence and matrix sensitivity in mixed-metal solutions; for *Galdieria sulphuraria* (SAG 21.92), removal efficiency was below 9% at pH 2.5 and up to 86% at pH 5.0, with no particular selectivity for REEs over Cu and Cu sorption 1.7–10× higher (molar basis) than any REE under tested conditions.
- Chemical separation benchmarks demonstrate both high throughput and high purity; multiple sources describe solvent extraction as the most useful/most successful industrial REE separation method and report that it can produce up to 99.99% purity material, with equilibrium for some systems reached within minutes.
- Commercial magnet recycling momentum in the cited sources is led by direct recycling and hydrometallurgical flowsheets; examples include Heraeus Remloy's production start (600 t/y, rampable to 1,200 t/y) and Cyclic Materials' investments and agreements targeting magnet production by-products and end-of-life magnet-containing equipment.

Methods

This report synthesizes the provided raw dataset of quoted excerpts spanning peer-reviewed papers and technical descriptions, alongside company websites and press releases included in the inputs. Biological performance claims were drawn from sources explicitly describing biosorption or bio-enabled recovery tested on either (a) real magnet-derived leachates (e.g., Nd solutions derived from real Nd–Fe–B magnets) or (b) mixed-metal solutions used to probe competition, pH sensitivity, and selectivity limitations. Chemical benchmark claims were extracted from sources describing hydrometallurgical processing, solvent extraction systems (D2EHPA, PC88A/HEHEHP, Cyanex series, TODGA, ionic liquids), leaching/precipitation routes, and reported commercial deployment statements where present. Company landscape statements were restricted to claims directly supported by the provided excerpts (technology descriptors, capacities, facility locations/status, funding/partnership statements when quoted).

Biological Technologies

The biological approaches represented in the provided sources fall into three main categories: (1) biosorption using microalgae/cyanobacteria/macroalgal biomass (often dried/lyophilized), (2) engineered cell-surface binding systems (notably LBT-displayed *E. coli*) adapted to column operation via hydrogel encapsulation, and (3) biofilm-based biomineralization concepts that precipitate REEs as phosphate minerals. The dataset includes magnet-relevant validation for *Chlorella* (real NdFeB-derived leachates) and for engineered *E. coli* (end-of-life NdFeB leachate), while many other biological studies are demonstrated on non-magnet or synthetic matrices, which remain relevant for understanding kinetics, pH windows, and competitive adsorption phenomena.

Microalgae and algae-derived biosorbents

Evidence for algae-based biosorption includes both magnet-derived leachate tests and non-magnet real leachates (e.g., phosphogypsum leachate), as well as synthetic mixed-metal systems that probe competition and selectivity limits. Across these systems, pH is repeatedly a key control lever, and some biosorbents show fast equilibration (minutes) while others require tens of minutes to hours, depending on organism and format.

Chlorella. Dried green microalgae (*Chlorella vulgaris*) were used to remove Nd from a mixed leachate derived from neodymium magnets, with maximum Nd uptake q = 157.21 mg/g at pH 5 (biosorbent dosage 0.5 g/L; initial Nd 250 mg/L; 35 ℃) in batch mode, and the work notes slow kinetics that reduced continuous-mode capacity relative to batch assays. A second *C. vulgaris* study emphasizes real-waste origin by stating that the Nd aqueous solution was derived from real Nd–Fe–B magnets, prepared by dissolving roasted sample in concentrated nitric acid, and reports a Langmuir qmax of 188.68 mg/g at 35 ℃ and pH 5 and pseudo-second-order kinetics with equilibrium within 30 minutes (with >90% uptake occurring before the slower approach to equilibrium). This study also reports pH sensitivity (tests at pH 3–5, with optimum at pH 5 at 21–35 ℃; and at 50 ℃ a maximum experimental uptake at pH 4 is reported in the excerpt).

Spirulina. Two *Spirulina platensis* strains (LEB-18, LEB-52) were used as biosorbents for Nd removal, with maximum biosorption capacities reported as 72.5 mg/g (LEB-18) and 48.2 mg/g (LEB-52) at 298 K, and removal percentage reported to reach 98.5% for LEB-18 and 77.5% for LEB-52 in the cited experiments. The same work reports kinetics (equilibrium around 80 minutes; t0.95 about 65–70 minutes) and pH dependence (pH evaluated 1.0–6.0; biosorption favored at pH 6; pHpzc values 5.62 and 5.81 for the two strains). Although not magnet waste, the strains were tested in real phosphogypsum leachate containing multiple REEs and were reported to remove more than 80% of Nd from that leachate even under acidic conditions. Biomass production scale is indicated by cultivation of LEB-52 in 450 L open outdoor photobioreactors (even though adsorption tests were at 50 mL scale in the excerpt).

Galdieria. In a mixed-metal system (Cu, La, Nd, Dy at 5 ppm each), raising initial pH to 5.0 increased REE sorption to 1.46–2.40 mg/g, with removal efficiency reported as below 9% at pH 2.5 and up to 86% at pH 5.0, and sorption at pH 1.5–2.5 (growth optimum) described as poor but improved up to 24-fold at pH 5.0 in phosphate-free conditions. However, the same study reports no particular selectivity toward REEs over Cu and Cu sorption 1.7–10× greater than any REE (molar basis) under tested conditions, which is a red flag for treating Cu-bearing industrial wastewaters without additional selectivity strategies. Operational constraints are also highlighted: biosorption should occur separately from growth to avoid REE phosphate precipitation, phosphate-limiting conditions are recommended, and no biosorption was observed under nutrient-replete conditions in the cited work. Another cited *G. sulphuraria* report states that Nd(III), Dy(III), La(III), and Cu(II) were recovered with >90% efficiency at 0.5 ppm and suggests that at pH 1.0–1.5 lanthanoid ions were collected more efficiently into cell fractions than Cu(II), implying a potentially different selective operating window in strongly acidic systems for lanthanoid/Cu separation (though this is not resolved into a magnet-waste demonstration in the excerpt).

Cyanobacteria and other phototrophs. A screening study reports that *Nostoc* sp. 20.02 showed the highest maximum adsorption capacity in the tested set (84.2–91.5 mg/g), and equilibrium was reached within 5 minutes, supporting the possibility of very rapid adsorption kinetics for some biomasses. The same work reports strong competition effects: adsorption capacity for cerium in mixed-metal solutions was significantly lower than in single-element solutions, and at 4 mM competitor metal concentration, other elements "seemed to replace cerium" as cerium binding capacity dropped across tested cyanobacterial biomasses. The study also notes that isolating single target elements via biosorption remains challenging due to surface heterogeneity and functional group complexity, and that adsorption properties can be impaired over a few cycles, which together define major scale-up barriers for single-REE separation using raw biomass.

A separate screening of 19 biological species (17 microalgae, one yeast, one moss) reports Nd biosorption qM values ranging from 0.26 mmol/g (*Chlamydomonas reinhardtii*) to 0.74 mmol/g (*Physcomitrella patens*), with kinetics reaching qM after 30 minutes for *C. reinhardtii* and 180 minutes for *P. patens* in the cited measurements. A pH study (pH 1–6) reports *P. patens* reached its maximum around pH 3 (0.72 ± 0.03 mmol/g) and remained nearly constant up to pH 6, while *Calothrix brevissima* and *Chlorella kessleri* increased with pH and reached highest values at pH 6, and cultivation scales are described from 0.1 L shake flasks to 15 L airlift reactor in the cited excerpt.

Macroalgae residues and composites. Ulva-derived biomass (including waste from Ulva biorefining) is described as a biosorbent for La, Nd, and Dy, where NaOH-assisted ulvan extracted biomass achieved recovery efficiencies of 90.71% (La), 95.12% (Nd), and 93.83% (Dy), and desorption using HCl and EDTA gave stable adsorption/desorption efficiency above 40% up to six cycles in the cited results. A composite sorbent based on *Fucus vesiculosus* alga, polyethylenimine, and calcium alginate is described, with maximum sorption capacity of 0.86 mmol/g at pH 4 and regeneration using 0.1 M HCl requiring two hours to recover >90% of total sorbed metal in the cited excerpt.

The table below consolidates the most magnet-relevant and decision-useful algae/cyanobacteria performance signals from the provided excerpts.

Biosorbent	REEs	Evidence type	Performance signals	Scale signals and constraints
Chlorella vulgaris (dried)	Nd	Mixed leachate derived from neodymium magnets	Max uptake 157.21 mg/g at pH 5 (0.5 g/L; Nd 250 mg/L; 35 ℃)	Continuous mode described as successful, but slow kinetics noted and continuous details not quantified in excerpt
Chlorella vulgaris	Nd	Nd solution derived from real Nd–Fe–B magnets (HDD); roasted sample dissolved in conc. HNO3	Langmuir qmax 188.68 mg/g at pH 5, 35 ℃; pseudo-2nd order; equilibrium within 30 min; >90% uptake early	Experiments in 500 mL flasks; selective leaching left Fe/B/Dy/Co below detection (selectivity vs Fe/Co not established)
Spirulina platensis (LEB-18/LEB-52)	Nd	Aqueous effluent + real phosphogypsum leachate	Capacity 72.5 mg/g (LEB-18) and 48.2 mg/g (LEB-52); removal up to 98.5% (LEB-18); equilibrium ~80 min; favored at pH 6	Biomass cultivated in 450 L outdoor photobioreactors; adsorption tests at 50 mL in excerpt; functional groups degrade above 40 ℃ (constraint)

Biosorbent	REEs	Evidence type	Performance signals	Scale signals and constraints
Galdieria sulphuraria (SAG 21.92)	La, Nd, Dy (+Cu)	Mixed-metal synthetic solutions	Removal <9% at pH 2.5 vs up to 86% at pH 5.0; no REE selectivity over Cu (Cu 1.7–10× higher mol sorption)	Phosphate/nutrient conditions strongly affect feasibility; phosphate-limiting recommended; no biosorption under nutrient-replete conditions
Cyanobacteria (e.g., *Nostoc* sp. 20.02)	Ce, Nd, Tb, La	Screening + competition tests	Max adsorption 84.2–91.5 mg/g; equilibrium within 5 min; mixed-metal competition can reduce/replace binding	Single-element isolation described as challenging; adsorption impairment over cycles noted generally
Ulva-derived biosorbent	La, Nd, Dy	Biomass/waste biosorbent	Recovery 90.71% La, 95.12% Nd, 93.83% Dy; HCl/EDTA desorption >40% up to 6 cycles	Optimized to perform at pH 7 in cited excerpt (potentially relevant to near-neutral manufacturing rinses)

Bacteria and engineered organisms

The bacterial evidence in the dataset includes engineered cell-surface binding systems for REE-selective capture, native bacterial biosorption with fast adsorption/desorption, and biofilm-based biomineralization concepts. The strongest magnet-specific validation in the provided sources is the engineered LBT-displayed *E. coli* platform tested on NdFeB leachate, including packed-bed breakthrough experiments, regeneration, and multi-cycle testing.

Engineered *E. coli* LBT platform. An *E. coli* strain engineered to display lanthanide binding tags (LBTs) on the cell surface was encapsulated within a permeable PEGDA hydrogel to create "microbe beads" for REE extraction. To test efficacy on an end-of-life feedstock, NdFeB magnet scraps from speakers of mobile devices were leached with sulfuric acid, and initial small-scale (1.5 mL) batch biosorption assays reported that at optimal cell density, 91% of REEs were extracted at 95% purity from the NdFeB magnet leachate. A bench-scale breakthrough experiment used a 1 × 100 cm (77 mL) microbe bead column and demonstrated effective Nd extraction at flow velocities up to 3 m/h at pH 4–6, and at the REE breakthrough point, REEs comprised 97% of the adsorbed metals by mass, while non-REE metal adsorption was minimal with transition metal breakthrough within a single bed volume.

This platform includes unusually explicit regeneration and cycle-life data for biosorption in the provided dataset: ~94% Nd desorption was achieved with 10 mM citrate (pH 6) within 3 bed volumes, and adsorption capacity retention was 85% after nine consecutive adsorption/desorption cycles. Kinetically, Nd adsorption increased rapidly over the first 10 minutes, reaching ~90% of the calculated adsorption maximum, and near complete Nd recovery was achieved within 5 minutes; free LBT-displayed *E. coli* showed even faster behavior with >80% of adsorption maximum and

near complete desorption within 1 minute in the cited comparison. The excerpts also document pH effects on bed capacity, with capacity reduced from 2.6 mg/g (pH 6) to 1.8 mg/g (pH 5) and 1.1 mg/g (pH 4), and a significant reduction in REE adsorption capacity below pH 4 in the cited observations.

Native bacterial biosorption. Biomass from the magnetotactic bacterium *Magnetospirillum magneticum* AMB-1 was evaluated for La(III) removal and recovery, reaching maximum uptake of 37.2 mg/g at 21 ℃ and pH 6 with rapid adsorption/desorption times of 8 minutes, and it could be reused for at least five rounds of La3+ adsorption/desorption in the cited work. The excerpt highlights an operational drawback: the need for biomass removal after adsorption can be expensive and energy intensive, a consideration that can dominate practical costs unless immobilization or alternative solids-handling strategies are used.

Biofilm biomineralization. A biofilm-based route is described in which enzymatically generated inorganic phosphate precipitates REEs as phosphate biominerals, described as highly effective and enabling rapid recovery onto immobilized bacterial biofilm at high flow-through rates in the cited excerpts. However, the same source states that the process requires addition of an organic phosphate substrate (e.g., glycerol 2-phosphate, G2P), and explicitly notes that G2P is expensive, precluding large-scale use, which creates a clear reagent-economics barrier unless cheaper substrates or recycling are implemented.

Biofilm Nd recovery studies. One cited effort reports examining microbial recovery of Nd using biofilm of Serratia sp. N14 and aimed to establish tolerance to low pH or high salt concentrations and develop semi-continuous biofilm preparation, but quantitative adsorption/recovery performance is not present in the provided excerpt.

The table below summarizes the bacterial systems with quantitative signals and indicates where magnet-relevant validation exists in the dataset.

System	Magnet relevance	Performance signals	Regeneration and durability	Key constraints
LBT-displayed *E. coli* microbe beads (PEGDA hydrogel)	Tested on sulfuric-acid leachate from end-of-life NdFeB magnet scraps (mobile device speakers); 77 mL packed-bed breakthrough experiments	91% REE extraction at 95% purity in 1.5 mL batch assays; REEs 97% of adsorbed metals at breakthrough; effective at flow velocity up to 3 m/h at pH 4–6; bed capacity ~2.6 mg/g (pH 6), reduced at lower pH	~94% Nd desorption with 10 mM citrate; 85% capacity retained after 9 cycles	Capacity penalty vs free cells noted via cited comparison (25.7 mg Nd/g dry cells); batch-processing constraints of many biosorbents motivate column formats
Magnetospirillum magneticum AMB-1 biomass	Not magnet-leachate validated in excerpt (La system)	Max uptake 37.2 mg/g (La3+) at pH 6, 21 ℃; rapid adsorption/desorption 8 min; usable for ≥5 rounds	Reuse stated for ≥5 rounds (no % loss reported in excerpt)	Biomass removal after adsorption described as expensive/energy intensive

System	Magnet relevance	Performance signals	Regeneration and durability	Key constraints
Biofilm phosphate biomineralization	Not magnet-waste validated in excerpt	Rapid recovery onto immobilized bacterial biofilm at high flow-through rates (qualitative in excerpt)	Not quantified in excerpt	Requires organic phosphate donor; G2P cost precludes large-scale use (as stated)

Fungi

The provided excerpts do not include quantitative fungal bioleaching results (e.g., *Aspergillus*/*Penicillium* organic-acid leaching of NdFeB powders), so fungal performance benchmarking and head-to-head comparisons to mineral acid leaching cannot be supported from this dataset segment.

Chemical Benchmarks

The incumbent chemical technology stack for magnet-related REE recovery is largely hydrometallurgical, with strong acid leaching to dissolve REEs from solid magnets and solvent extraction (SX) and/or chromatography steps used for group separation and high-purity splitting. Multiple sources characterize hydrometallurgy as the most common method to extract REEs (primary and secondary) and identify solvent extraction as the most useful/most successful industrial separation method for REEs from chemically complex aqueous matrices, including claims that SX can yield up to 99.99% purity materials in cited sources.

Solvent extraction

Solvent extraction is repeatedly described as the best/most useful/most successful large-scale REE separation technology in the provided excerpts, and it is presented as the dominant industrial method for separating REEs from complex leachates. D2EHPA is highlighted as a versatile extractant due to stability, kinetics, loading/stripping properties, and commercial availability.

For NdFeB-related leachates, one cited set of conditions reports that 100% of REEs were transferred to the organic phase at 0.9–1.2 M D2EHPA in hexane and octane, and another excerpt adds that under these conditions no Co or B was extracted, indicating potential impurity rejection for specific magnet-leachate compositions. Stripping can be highly efficient, with 100% stripping reported at ≥2 M HCl in the cited experiments. Selectivity can be tuned, with the highest separation factors between Dy and Nd achieved at 0.3% D2EHPA in hexane in one cited excerpt, where around 50% of Nd/Pr were extracted while around 99% of Dy/Gd were extracted, and typical separation factors between adjacent REEs using D2EHPA are described as around 2.5 in cited references.

Process kinetics can be fast: equilibrium is reported reached within ~3 minutes for all diluents except cyclohexanone and 1-octanol (5 minutes) in one excerpt, which illustrates why SX circuits can be high-throughput relative to many biological methods. Multi-stage equipment is used for sharp separations, including an 11-stage mixer-settler achieving near theoretical separation in a cited continuous process demonstration and a seven-stage process reporting 99.5% Sm organic recovery and 80% Nd aqueous recovery that was validated in a continuous trial on a mini-pilot scale in the cited excerpts.

Beyond D2EHPA, other extractants and systems are represented. Cyanex 923 is associated with extremely high Dy/Nd separation (separation factor larger than 6000 at 0.5 M) in the cited excerpt, and a McCabe–Thiele diagram indicated complete Dy extraction could be achieved via a 3-stage counter-current process in that cited work. TODGA is described as extracting REEs selectively and completely as a group at 0.1 M with stripping over 98% in one step using water in the cited results, and a real-leachate extraction study is described as performed in small-scale (3.5 mL) vials, underscoring that some high-performance systems remain at small experimental scales in the provided dataset.

Ionic liquids are described as greener options (low vapor pressure, high stability, less flammability) and include specific magnet-related separation workflows. The phosphonium ionic liquid trihexyl(tetradecyl)phosphonium nitrate is cited for separation of Nd(III), Dy(III), and Co(II), producing Nd2O3 (99.6%), Dy2O3 (99.8%), and CoO (99.8%) with reported Nd/Co and Dy/Co separation factors (αNd/Co 590; αDy/Co 500). Another ionic-liquid system, [P66614][Cy272], is cited as removing 100% of iron from NdFeB leach liquor with minimal co-extraction of REEs (~4%), which is directly relevant to the magnet-leachate challenge of separating REEs from abundant Fe.

Acid leaching and precipitation

Acid leaching is a foundational unit operation for NdFeB recycling hydrometallurgy; the provided excerpts state that REEs are normally leached from NdFeB magnet waste using strong mineral acids such as HCl, HNO3 and H2SO4, and also indicate that a variety of inorganic and organic reagents have been tested as lixiviants in magnet-leaching studies. One excerpt reports that leaching percentages for Nd and Dy were above 90% for higher HCl-to-lanthanide ratios, and another notes that inorganic acids dissolved all elements with yields over 83% after 5 hours, while also stating that selectivity coefficients for these acids were close to one (no selectivity observed), emphasizing why downstream separation is required after mineral-acid leaching.

Co-leaching of iron is a recurring issue; one excerpt reports about 35% of iron in feed co-leached with REEs upon leaching of thermally oxidized products, and another explicitly notes the need for oxidation pretreatment to improve selectivity between Nd and Fe. As a counterexample, a deep eutectic solvent (DES) system is cited achieving Nd leaching ratio up to 97.63% while Fe is less than 0.435% and the separation factor can exceed 9000, demonstrating that selective leaching can dramatically reduce downstream purification burden when it works on target feedstocks.

Precipitation is a standard recovery and concentration step. Oxalic acid precipitation is cited as a route for rare earth precipitation, and one excerpt states that at an n(oxalic acid)/n(Nd) of 1.5 precipitation increased linearly toward 100%; another states oxalic acid forms very strong water-insoluble complexes with rare earths, and a separate excerpt states oxalate precipitation stripping can achieve 100% stripping using stoichiometric oxalic acid. A pH-controlled precipitation route is also cited: neodymium precipitated from leached solution by pH control to pH 0.6 using NaOH, and the final precipitate composition (WD-XRF) showed 26.36% Nd and 8.13% Pr with Fe as low as 0.14% Fe in the cited excerpt, indicating substantial Fe rejection in the precipitated product under those conditions.

Ion exchange and extraction chromatography

The provided excerpts describe ion exchange/resin approaches as used for metal extraction while noting fundamental difficulty: separation of REEs in ultrapure form by resins is difficult due to the similarity of chemical properties across REEs. One excerpt describes efforts to improve resin selectivity using D2EHPA, reporting that extraction of Nd was marginally higher than Pr and that stripping of Nd was significantly higher than Pr in both HCl and H2SO4 media in the cited results.

A more process-integrated approach is also cited: a REE-loaded H2SO4 strip solution was separated into a high-purity Nd and Pr fraction from separate Sm and Dy fractions in a single stage using a D2EHPA-impregnated column and

H2SO4 gradient elution, with Pr recovered with >92% Nd in a fourth fraction and combined Nd+Pr purity >98% in the cited excerpt, and Sm and Dy recovered in separate fifth and sixth fractions at specified H2SO4 eluate concentrations.

Electrochemical and membrane methods

Electrochemical and membrane approaches are represented mostly via pilot/demonstration descriptions and enabling system statements in the provided excerpts. REE4EU is cited as integrating high temperature electrolysis (HTE) and ionic liquid extraction (ILE) and as having successfully treated several tons of in-process wastes and end-of-life products containing REEs, recovering almost one hundred kilograms of rare earth alloys, with the pilot described as in operation in the cited material. Nth Cycle is cited as commercializing patented "Electro-Extraction" technology that replaces conventional energy-intensive pyrometallurgy, although the excerpt does not provide magnet-specific performance figures.

For membrane systems, one excerpt describes a "membrane-based, bench-scale system to extract strategic minerals," and another reports a recycling process using membrane-assisted solvent extraction that yielded a 95% rare earth recovery rate in the cited statement, positioning membranes as process intensification or preconcentration components in some flowsheets.

Benchmark snapshot table

The table below summarizes a compact set of chemical benchmark datapoints explicitly present in the excerpts, focusing on performance and selectivity signals relevant to magnet leachates.

Technology	Example datapoints in provided sources	Implications for magnet waste
Solvent extraction (D2EHPA)	100% REE transfer at 0.9–1.2 M D2EHPA (hexane/octane); 100% stripping at ≥2 M HCl; equilibrium reached within 3–5 min in cited kinetics	High-throughput, scalable separation; can be tuned for Dy vs Nd/Pr selectivity, but adjacent-REE SF often ~2.5 in cited references
Solvent extraction (Cyanex 923)	Dy/Nd separation factor >6000 at 0.5 M; complete Dy extraction achievable by 3-stage counter-current in cited McCabe–Thiele analysis	High separation intensity for Dy/Nd splitting (potentially fewer stages for targeted separations)
Ionic liquids	Nd2O3 99.6%, Dy2O3 99.8%, CoO 99.8%; αNd/Co 590, αDy/Co 500 in cited workflow; 100% Fe extraction with ~4% REE co-extraction in cited [P66614][Cy272] system	Promising impurity (Fe, Co) separations; many demonstrations remain small-scale in the excerpts
Selective leaching (DES)	Nd leaching ratio 97.63% with Fe <0.435% and separation factor >9000 in cited DES system	Upstream selectivity can radically simplify downstream separations, directly addressing Fe co-leach problem

Technology	Example datapoints in provided sources	Implications for magnet waste
Oxalate precipitation	n(oxalic acid)/n(Nd)=1.5 drives precipitation toward 100%; stoichiometric oxalic acid sufficient for 100% stripping in cited oxalate stripping statement	Robust group recovery/concentration step; typically followed by separation for individual REEs

Company Landscape

The company landscape in the provided sources includes (1) magnet-to-magnet direct recycling players (often avoiding full chemical disassembly), (2) hydrometallurgical recyclers producing oxides or intermediate products from magnet waste and end-of-life equipment, and (3) emerging biological and bio-inspired separation companies working on engineered microbes, engineered proteins, and bioleaching/bioseparation platforms. The excerpts provide the clearest commercial-scale signals for direct recycling and hydrometallurgical magnet recycling, while biological companies are presented as commercializing but generally earlier-stage in the cited materials.

Magnet recycling companies

HyProMag. HyProMag's core technology is described as patented Hydrogen Processing of Magnet Scrap (HPMS) technology to extract and demagnetise NdFeB alloy powders from magnets embedded in scrap and redundant equipment, with the HPMS process described as exposing products to hydrogen at atmospheric pressure and room temperature so magnets break apart into demagnetised powder that is easier to extract. HPMS is described as feedstock agnostic in the cited material, and one cited press release states the technology uses 88% less energy and reduces carbon emissions by 85% compared to conventional methods. A Benchmark Minerals source states HyProMag inaugurated the UK's first commercial rare earth magnet recycling and manufacturing facility (January) and reports capacity to produce up to 300 tonnes of NdFeB magnet blocks per year at full capacity, with validation by GKN and Toyota supporting OEM confidence in the cited statement. Other cited excerpts note HyProMag Germany is constructing a plant at Pforzheim and that HyProMag USA targets commissioning a Texas facility by mid-2027 (subject to permitting/financing), with an annual output figure of 750 metric tons of recycled sintered NdFeB magnets and 807 metric tons of associated co-products cited for the US project.

Cyclic Materials. Cyclic Materials states that its proprietary MagCycle and REEPure processes can extract REEs and claims recovery rates exceeding 98% in the cited press release excerpt, while also stating it recovers REEs from EV drive motors, hybrid transmissions, wind turbine generators, hard disk drives, MRI machines, and magnetic waste sourced from magnet manufacturers in cited materials. A cited source notes the company commissioned a commercial demonstration facility of the first stage of its process, and another excerpt states it is investing more than USD $82M to establish a rare earth recycling campus in McBee, South Carolina, with initial capacity to produce 600 tonnes of MREO per year and planned expansion to 1,800 tonnes in cited materials. The company also cites a 10-year exclusive agreement with VACUUMSCHMELZE (VAC) to recycle 100% of magnet production by-products in the cited press release excerpt, directly relevant to magnet manufacturing waste streams (swarf, by-products, off-spec materials).

Heraeus Remloy. Heraeus states it started production at Europe's largest recycling plant for rare earth magnets, and the facility is described as producing 600 tons/year of nanocrystalline rare earth magnetic powders from recycled end-of-life electronic devices, with capacity rampable to 1,200 tons/year in cited excerpts; the sources also cite waste

magnets from computer hard disks as a target feedstock. A cited trade source states the technology reduces CO2 consumption by 80% compared to primary production while achieving the same magnet quality in the cited statement, and the Heraeus excerpt notes the product can be used to produce bonded NdFeB magnets, hot formed NdFeB magnets, and rare earth hydrogen storage components.

Noveon Magnetics. Noveon is described as operating at a factory in San Marcos, Texas and recycling rare-earth magnets through a powder metallurgy process to produce its patented EcoFlux magnet, which can be made from mined rare earths or directly from recycled end-of-life magnetic materials in the cited excerpts. Noveon's 145,000 square foot San Marcos facility is described as commercially operational and cited text states it will be able to produce 2,000 tons of magnets under long-term supplier agreements; early customers cited include Nidec Motor and Eriez in the excerpted source.

Geomega Resources. Geomega describes developing and licensing closed-loop hydrometallurgical processes and states its process chemically disassembles magnets into elemental components within a closed-loop, zero-waste system in cited excerpts. The company claims >90% recovery and >99% purity and also claims >99% TREO purity for Nd, Pr, Dy, Tb in cited materials. A Saint-Hubert demonstration plant is described as under construction near Montréal with TRL 7–8 and designed to process 1.5 tonnes of magnet waste per day, financed by federal and provincial governments ($8M CAD) in the cited excerpts; the company also explicitly includes end-of-life magnets, production scraps and swarf from magnet machining as target feed streams in the cited material.

Biological metal recovery companies

REEgen. A Nature excerpt identifies REEgen as commercializing REE biomining technology, and the company states it uses engineered bacteria to recover critical metals from ores and unconventional feedstocks efficiently and economically with no harsh chemicals or high heat in the cited excerpt. A news excerpt states the startup uses genetically engineered bacteria to recover REEs and other critical minerals from industrial waste and references "BioREEcover" technology, and the same source states an aim to produce rare earth products for the U.S. supply chain by end of 2026 in the cited text. The Nature excerpt reports that the highest performing strain (*G. oxydans* ΔpstS, P112:mgdh) performs up to 73% better at REE-bioleaching than wild-type in the cited statement, providing a quantitative R&D performance signal (though not a magnet-waste demonstration in the excerpt). REEgen's cited milestones include receiving a $1.1M NSF SBIR Phase II award and winning a $150,000 prize in 2025 in the cited excerpts.

Alta Resource Technologies. Alta states its platform uses engineered proteins to selectively bind and separate metal ions and to separate REEs with "unprecedented selectivity," and it claims the target metal can be released from the proteins producing a high-purity (>99%) output in cited excerpts. The company's target feedstocks include end-of-life magnets in e-waste (including hard disk drives, cellphones, speakers), EV and wind turbine motors, MRI machines, defense equipment, and also mining tailings and coal ash in the cited text. Alta also reports backing by a set of venture capital firms and support from DARPA and DOE-associated programs in cited materials, indicating institutional validation and funding support (though the excerpt does not provide plant capacities or deployments).

BIOWEG. BIOWEG is described as developing bio-leaching of rare earth elements using precision fermentation and engineered biological systems, and one cited excerpt states its platform combines BIOWEG's bioacid production (from waste streams) for bioleaching with TU Berlin's peptide-based separation technology using column systems, implying a hybrid bioleach + selective capture strategy in development. The process is described as operating in water at ambient temperature without solvents or high heat, and BIOWEG is cited as operating a demonstration facility in Quakenbrück, Germany with pilot-scale fermentation and processing equipment (and a lab in Monheim on the Bayer Crop Science campus) in the cited excerpts. A cited source provides a lab throughput figure (0.5–2 L/hour) and also cites a processing duration of 72–168 hours for complete extraction cycles, which sets expectations

for bioleaching time scales relative to solvent extraction.

Established REE processors and recycling-adjacent players

Some large incumbents are included in the provided excerpts as context for processing hubs and supply chains. Lynas is described as operating a mine at Mt Weld in Western Australia and operating the world's largest single rare earth processing plant in Malaysia in cited excerpts, with plans for additional facilities including a separation facility in Texas supported by U.S. programs and a cited 2025 milestone of producing commercial quantities of dysprosium oxide outside China. MP Materials is described as working "mining through magnet manufacturing" in a cited U.S. supply chain report excerpt and is associated with a planned facility producing 1,000 tonnes of NdFeB magnets per year in cited text; Umicore is cited as partnering with BMW Group to establish a dedicated rare earth recycling facility in Germany in the cited market report excerpt.

Company comparison table

The table below provides a compact, citation-grounded comparison of selected companies from the provided excerpts, focusing on technology type, target streams, and scale signals.

Company	Technology	Target streams	Scale and status signals in provided excerpts
HyProMag	HPMS hydrogen processing (direct recycling of NdFeB alloy powders)	Magnets embedded in scrap/redundant equipment; end-of-life magnets and manufacturing scrap referenced in cited coverage	UK commercial facility referenced; capacity up to 300 t/y NdFeB magnet blocks at full capacity; Germany plant under construction; USA Texas facility targeted mid-2027; US project output 750 t/y recycled sintered NdFeB magnets (plus co-products)
Cyclic Materials	MagCycle + REEPure (two-stage physical + hydromet recycling in cited description)	EV motors, wind turbines, HDDs, MRI machines; magnetic waste from magnet manufacturers; agreement to recycle 100% VAC magnet production by-products	Claimed recovery rates >98%; commercial demonstration facility referenced; SC campus planned with 600 t/y MREO initial and 1,800 t/y expansion
Heraeus Remloy	Recycling plant producing nanocrystalline rare earth magnetic powders (chemical-free described in trade coverage excerpt)	End-of-life electronic devices and HDD magnets cited as feedstocks	Production started May 2024; 600 t/y capacity with ramp to 1,200 t/y in cited excerpts

Company	Technology	Target streams	Scale and status signals in provided excerpts
Noveon Magnetics	Powder metallurgy magnet-to-magnet recycling; EcoFlux magnets	Discarded items including EVs and hard drives; harvesting end-of-life magnetic materials cited	San Marcos, Texas facility commercially operational; cited plan to produce 2,000 tons of magnets under long-term supplier agreements; early customers include Nidec Motor and Eriez
Geomega Resources	Closed-loop hydrometallurgical chemical disassembly to elemental components (zero-waste claimed)	End-of-life magnets, production scraps, and machining swarf cited as targets	Demo plant under construction near Montréal (TRL 7–8) designed for 1.5 t/day magnet waste; claims >90% recovery and >99% purity and >99% TREO purity (Nd, Pr, Dy, Tb)
REEgen	Engineered bacteria biomining/bioleaching platform (commercializing stated)	Industrial wastes (slags, electronics, coal ash) cited; non-magnet specific in excerpts	R&D performance signal: engineered strain up to 73% better than wild-type at REE bioleaching; stated aim for US supply chain by end of 2026; NSF SBIR Phase II award cited
Alta Resource Technologies	Engineered protein binding/separation (lanmodulin-linked in company description)	End-of-life magnets (HDDs, cellphones, speakers), EV/wind motors, MRI/defense equipment; tailings/coal ash also cited	Claims high-purity (>99%) output; funding/support from VC firms, DARPA, DOE programs cited (no plant capacity in excerpt)
BIOWEG	Precision-fermentation bioacids + peptide-based column separation with TU Berlin cited	End-of-life smartphone/computer recovery and permanent magnet recycling from EV motors cited in secondary coverage excerpts	Demonstration facility in Quakenbrück cited; lab throughput 0.5–2 L/h and extraction cycles 72–168 h cited in excerpts

Executive Synthesis

The executable market for a biological REE recovery process from NdFeB magnet manufacturing waste is anchored in a compelling compliance-driven business model, with a critical and fleeting "design-in" window between 2024 and 2029. A significant buildout of non-Chinese magnet manufacturing capacity is underway in the US and EU, creating a time-sensitive opportunity to be specified into new plant designs as the default wastewater and sludge management solution. The primary value proposition is not REE commodity recovery, but rather avoided and reduced compliance costs for plant operators. These costs include high operational expenses for wastewater treatment and substantial, geographically variable fees for the disposal of hazardous metal-bearing sludge. Manufacturing waste streams, particularly grinding sludge and swarf, contain economically significant concentrations of Nd, Pr, and Dy (totaling up to 30-37 wt%) but are contaminated with high levels of iron, oils, and other additives that complicate treatment. Existing regulations in the EU (IED/BREF) and US (EPA Metal Finishing/Nonferrous Metals categories) impose stringent effluent limits for competing metals like chromium, nickel, and zinc, and enforcement actions carry penalties reaching tens to hundreds of thousands of dollars, plus mandatory upgrades. This creates a clear pain point for operators, making a compliance-first solution that reduces sludge volume, lowers its hazardous classification, and simplifies permitting a high-value proposition. The secondary upside from REE sales is a bonus, insulating the business model from commodity price volatility.

Plant Capacity Buildout

A significant expansion of NdFeB magnet manufacturing capacity is underway in the United States and Europe, driven by a strategic imperative to build non-Chinese supply chains. This buildout presents a critical "design-in" window for new waste treatment technologies. Integrating a novel REE recovery process during a new plant's design and commissioning phase (typically 2024-2029 for the announced projects) is substantially more efficient and less disruptive than retrofitting an existing, operational facility. Targeting engineering and environmental leads at these companies during their planning stages is the most executable beachhead strategy.

Key announced capacity additions include:

Company Name	Facility Location	Capacity (Tonnes/Year)	Commissioning Year	Investment Amount	Status (as of early 2026)
MP Materials	Fort Worth, TX, USA	~1,000	Production ramp began late 2025	$58.5M (tax credit)	Operational (Trial/Ramp-up)
E-VAC Magnetics (VAC Group)	Sumter, SC, USA	Not Specified	Late 2025	>$500M (project); $94.1M (DoD grant)	Under Construction
Neo Performance Materials	Narva, Estonia	2,000 (Phase 1); planned to 5,000	2025	€75M (Phase 1)	Operational
USA Rare Earth	Stillwater, OK, USA	~1,200 (initial); ramp to 4,800	2025 (initial batch); ramp by ~2026	>$100M	Initial Production
Noveon Magnetics	San Marcos, TX, USA	>2,000 (expansion target)	Operational; expanding	$215M (Series C financing)	Operational & Expanding

Company Name	Facility Location	Capacity (Tonnes/Year)	Commissioning Year	Investment Amount	Status (as of early 2026)
HyProMag USA	Dallas-Fort Worth, TX; South Carolina; Nevada	4,656 (total across 3 hubs)	2027-2029	Not specified	Announced / Pre-Feasibility
POSCO International / Star Group	Texas, Tennessee or Arizona, USA	3,000 (initial); ramp to 5,000	Not specified	Not specified	Site Selection / Planned
SGI Vina	Quang Nam Province, Vietnam	4,000-5,000	2025	$80M	Installing Equipment
GKN Powder Metallurgy	Radevormwald, Germany (pilot)	4,000 (planned across EU/NA)	Early 2025 (pilot)	Not specified	Pilot Launched

Waste Generation & Chemistry

The feasibility of a biological REE recovery process is critically dependent on the chemistry of the manufacturing waste streams. Academic and patent literature provides insights, but there is a significant distinction between data from true industrial matrices versus lab-prepared solutions. Industrial streams are complex mixtures containing high concentrations of competing metals and organic additives.

- **Sintering Residues & Grinding/Cutting Sludge/Swarf**: These solid and slurry wastes are the most REE-rich streams. Studies on true industrial samples show total REE content can be as high as 30-40 wt%. Neodymium (Nd) is the dominant REE, often ranging from 12-24 wt%, with praseodymium (Pr) and dysprosium (Dy) also present in smaller but significant amounts. However, iron (Fe) is the most significant competing metal, often exceeding 50-65 wt%. These streams are also contaminated with organics from cooling and lubricating fluids, which can constitute over 50% of the slurry's mass.
- **Electroplating Effluent & Wastewater**: Data on REE concentrations in electroplating wastewater from magnet manufacturing is scarce. The primary challenge in these streams is the presence of complexing agents, surfactants, and other additives used in plating baths, alongside competing metal ions from the plating process itself (e.g., Ni, Cu, Zn). While not directly from magnet manufacturing, analogous data from the metal finishing industry shows these wastewaters require extensive treatment to remove heavy metals and organics.
- **Filter Cake**: This waste stream results from the dewatering of various slurries. Its composition will reflect the upstream process it derives from, typically concentrating the solid metal particles (including REEs and Fe) and retaining some of the liquid-phase contaminants.

The table below summarizes representative data points for real industrial waste streams. Note the distinction between lab-prepared leachates and analyses of actual industrial materials.

Metric	Value	Unit	Year	Geography	Waste stream	Conditions	Evidence level	Source
REE Concentration (Total)	up to 30	wt%	2023	Not specified (academic study)	Diamond cutting and grinding waste	As-received machining waste	Commercial sample, lab analysis	
Nd Concentration	20.64	wt%	2016	China (patent)	NdFeB sludge (pre-treated powder)	XRF analysis	Commercial sample, lab analysis	
Nd Concentration	30.2	wt%	2023	Not specified (academic study)	Diamond cut machining residue	XRF analysis	Lab-generated industrial analog	
Pr Concentration	6.46	wt%	2016	China (patent)	NdFeB sludge (pre-treated powder)	XRF analysis	Commercial sample, lab analysis	
Dy Concentration	2.59	wt%	2016	China (patent)	NdFeB sludge (pre-treated powder)	XRF analysis	Commercial sample, lab analysis	
Fe Concentration	67.31	wt%	2016	China (patent)	NdFeB sludge (pre-treated powder)	XRF analysis	Commercial sample, lab analysis	
Fe Concentration	66.4	wt%	2023	Not specified (academic study)	Grinding residue	XRF analysis	Commercial sample, lab analysis	
Organics Content	52.5	wt%	2016	Global (estimate)	Acoustic magnet swarf slurry	As-received slurry	Market/Policy Estimate	

Compliance Economics

The primary economic driver for adopting a new REE recovery technology in magnet manufacturing is the high cost of compliance with environmental regulations for wastewater and hazardous sludge. This "avoided cost" model provides a revenue floor independent of volatile REE commodity prices.

- **Wastewater Treatment Costs**: For analogous industries like metal finishing, operating costs for wastewater treatment can range from approximately $1.37 to $4.88 per cubic meter, depending on the technology and influent characteristics. These costs encompass energy, chemicals, and labor for processes designed to remove heavy metals to comply with discharge permits.
- **Sludge Disposal Costs**: Sludge from treating metal-bearing wastewater is often classified as hazardous, incurring significant disposal fees. Costs vary widely by geography:
  - **Europe**: Median hazardous waste management costs were reported at €238 per tonne in a 2013 survey. Landfill taxes for hazardous waste can be substantial, with rates in 2021 ranging from €22/tonne in Slovenia to over €100/tonne in Belgium.
  - **United States**: Hazardous waste landfilling costs (with stabilization) from a 1999 EPA report were around $190/ton (adjusted for inflation from the original $185/ton). More recent (2021) data from California shows average disposal fees of $310 per ton for restricted hazardous waste.
  - **Asia**: Data from Tokyo suggests industrial sludge disposal costs reached about $300/ton, while estimates from India place hazardous sludge disposal at INR 15,000-25,000 per ton (approx. $180-$300/ton).
- **Permitting and Monitoring**: Beyond direct treatment and disposal, significant overhead exists for permitting, monitoring, and reporting. EPA documents from the 1990s estimated annual recordkeeping costs at $25,000 and RCRA permit acquisition from $250,000 to $500,000. A 2025 estimate from India suggests annual documentation and compliance overhead for a medium-sized unit could be INR 50,000-100,000.

These costs represent a direct, recurring operational expense for magnet manufacturers. A technology that can reduce the volume of sludge, lower its hazardous classification (enabling cheaper disposal), and simplify wastewater treatment has a clear, quantifiable value proposition based on these avoided costs.

Regulatory & Permitting Map

The regulatory landscape for NdFeB magnet manufacturing waste is governed by a patchwork of regulations for industrial wastewater, hazardous waste, and sector-specific rules for metal processing. There are no explicit effluent discharge limits for Neodymium (Nd) or Dysprosium (Dy) in the primary regulatory frameworks for the EU or US.

European Union

The **Industrial Emissions Directive (IED) 2010/75/EU** is the core legislation. It requires industrial installations to operate with a permit based on **Best Available Techniques (BAT)**.

- **BREFs and BAT Conclusions**: The relevant BAT Reference Documents (BREFs) are for **Non-Ferrous Metals Industries (NFM)** and **Surface Treatment of Metals and Plastics (STM)**. These documents contain legally binding BAT Conclusions that national authorities must use to set permit conditions, including **BAT-Associated Emission Levels (BAT-AELs)**.
- **Effluent Limits**: The NFM and STM BREFs provide BAT-AELs for common metals like Zinc (0.05-1 mg/l), Nickel (0.01-0.2 mg/l), and Copper, but **Nd and Dy are not listed**. Limits for REEs would likely be set on a case-by-case basis by national permit writers, considering local environmental quality standards.
- **Sludge Classification**: Sludge is classified as hazardous or non-hazardous based on the EU Waste Framework Directive and the List of Wastes. The presence of heavy metals above certain thresholds will typically render sludge hazardous, requiring specialized and more expensive disposal routes.

United States

The **Clean Water Act** governs wastewater discharges through the **National Pollutant Discharge Elimination System (NPDES)** program.

- **EPA Effluent Guidelines**: Magnet manufacturing facilities are likely to be regulated under the **Metal Finishing Point Source Category (40 CFR Part 433)** or the **Nonferrous Metals Manufacturing Point Source Category (40 CFR Part 421)**.
- **Effluent Limits**: These regulations set technology-based effluent limits for pollutants such as Cadmium, Chromium, Copper, Lead, Nickel, and Zinc. As in the EU, specific limits for **Nd and Dy are Not found** in these national standards. Limits would be established in a facility's NPDES permit based on state water quality standards or on a Best Professional Judgment (BPJ) basis by the permit writer.
- **Sludge Classification**: Sludge from wastewater treatment in this sector (e.g., F006 waste code for electroplating sludge) is often listed as a hazardous waste under the **Resource Conservation and Recovery Act (RCRA)**, requiring cradle-to-grave management and disposal in a licensed hazardous waste facility.

China

China's regulatory framework is driven by national standards issued by the Ministry of Ecology and Environment (MEE).

- **Applicable Standards**: The **Emission Standard of Pollutants for Rare Earths Industry (GB 26451-2011)** is the most relevant sector-specific regulation. For electroplating, the **Emission standard of pollutants for electroplating (GB 21900-2008)** would apply.
- **Enforcement**: China has a permit system for pollutant discharge, and enforcement has been strengthening, with a focus on key industries like rare earths.

Market Mechanics

The target customer for an on-site REE recovery solution is the **Plant Manager** or **Environmental, Health, and Safety (EHS) Manager** at an NdFeB magnet manufacturing facility. The payment for such a system or service comes from the plant's operational expenditure (OPEX) or capital expenditure (CAPEX) budget, specifically from budget lines allocated to **wastewater treatment** and **hazardous waste management**.

Buying Triggers:

1. **New Plant Construction**: The primary trigger is the design phase of a new manufacturing plant. This "greenfield" opportunity allows the technology to be designed-in from the start, becoming the baseline for the facility's environmental compliance infrastructure.
2. **Regulatory Enforcement Action**: A notice of violation, fine, or consent order from a regulator (e.g., EPA, EU Member State authority) for exceeding discharge limits or improper sludge disposal creates an immediate and urgent need for a new or improved compliance solution.
3. **Permit Renewal/Modification**: When a plant's operating permit is up for renewal, regulators may impose stricter limits, creating a need to upgrade treatment systems.
4. **Cost Reduction Mandates**: Corporate or plant-level mandates to reduce operational costs can trigger a search for more efficient waste management solutions, especially if disposal costs are escalating.

Procurement Motion: The sales process is a technical, value-based sale, likely involving:

1. **Initial Assessment & Scoping**: Analyzing the specific waste streams of the target facility (volumes, chemistry, variability).
2. **Pilot Demonstration**: A pilot-scale demonstration, potentially on-site, to prove the technology's efficacy on the plant's actual waste matrix. This is crucial for de-risking the solution for the customer.
3. **Techno-Economic Proposal**: A detailed proposal quantifying the value proposition, focusing on avoided costs (e.g., "Our system will save you $X per year on sludge disposal and $Y on wastewater treatment chemicals"), payback period, and compliance assurance.
4. **Contracting**: Contracts could be structured as a technology sale (CAPEX), a lease model, or a service agreement (e.g., per cubic meter of wastewater treated).

Competitive Context

The primary competition is the customer's **status quo** for waste management. For a new technology to be adopted, it must offer a compelling advantage over these existing methods.

Current Practices & Substitutes:

1. **Conventional Wastewater Treatment**: Most facilities use standard chemical precipitation (e.g., hydroxide precipitation), flocculation, and clarification to remove heavy metals from wastewater. This process is well-understood and effective for common regulated metals but generates large volumes of metal hydroxide sludge.
2. **Off-site Disposal**: The generated sludge, often classified as hazardous, is dewatered (e.g., using a filter press) and then transported by a licensed hauler to a specialized landfill or incinerator. This is a purely cost-based service with no value recovery. The willingness to pay for an alternative is directly tied to reducing these disposal fees.
3. **No On-site Recovery**: For the most part, REEs in manufacturing waste streams are not currently recovered on-site. The value of the contained REEs is lost when the sludge is disposed of.

Impact on Willingness to Pay:

- A new solution does not need to be cheaper than conventional treatment if it significantly reduces the much larger cost of sludge disposal.
- The key is to shift the sludge from a hazardous to a non-hazardous classification or to dramatically reduce its volume, thereby lowering disposal costs.
- The value of the recovered REEs is a secondary benefit that can improve the overall economics but should not be the primary driver of the sale. The business case must stand on the "avoided cost" of compliance alone.

Enforcement Cases

Real-world enforcement actions provide quantitative evidence of the financial and operational risks of non-compliance, directly influencing a potential customer's willingness to pay for a robust compliance solution. The following cases from the metal finishing, electroplating, and rare earth processing sectors illustrate these risks.

Jurisdiction	Year	Violation Type	Penalty/Cost	Industry Sector	Relevance
US EPA Region 9 (California)	2015	Untreated wastewater discharge; hazardous waste violations	**$142,200** (Anaplex)	Metal Finishing	Direct analog for electroplating effluent violations; demonstrates significant penalty size for combined CWA and RCRA failures.
US EPA Region 5 (Ohio)	2021	NPDES permit limit exceedances (TSS, chlorine)	**$55,300**	Ferroalloy Production	Demonstrates enforcement for common effluent parameter violations from a metal processing facility, relevant to general wastewater compliance.
US EPA Region 9 (California)	2012	Failure to treat wastewater to federal standards	**$74,000**	Metal Plating	Shows significant fines for pretreatment violations in a direct analog industry (electroplating).
US EPA Region 3 (Pennsylvania)	2024	Failure to file TRI reports for lead	**$31,653**	Industrial Plating	Highlights that reporting and documentation failures, not just discharge violations, carry substantial financial penalties.

Jurisdiction	Year	Violation Type	Penalty/Cost	Industry Sector	Relevance
State of California	2019	Order to investigate PFAS discharge	**$20k-$100k** (est. investigation cost)	Chrome Plating	Illustrates high costs of investigation and monitoring imposed by regulators, even without a direct fine, for emerging contaminants.
US EPA (Seattle, WA)	2008	Failure to report hazardous waste spill	**$60,000**	Electroplating	Quantifies the penalty for procedural failures like reporting, separate from the cost of the spill itself.

These cases underscore that the "cost of non-compliance" includes not only fines but also mandatory capital upgrades, extensive monitoring requirements, and significant legal and administrative overhead. This strengthens the business case for a solution that offers "compliance-as-a-service" and de-risks a plant's operations.

REE Pricing

The value of recovered Rare Earth Elements (REEs) should be treated as a secondary, optional upside to the primary compliance-driven business model. The REE market is notoriously volatile, and a business case dependent on REE spot prices is inherently risky. Nevertheless, recovering these critical materials can provide a significant additional revenue stream.

Price History and Volatility:

- **Neodymium (Nd)** and **Praseodymium (Pr)**, often priced as a combined oxide (NdPr), have seen significant fluctuations. For example, historical data shows prices for NdPr oxide ranging from approximately 360,500 CNY/T in early 2020 to a peak of over 1,500,000 CNY/T in early 2022 before settling to around 652,500 CNY/T in late 2023. In May 2024, NdPr oxide was assessed at approximately $51/kg in China.
- **Dysprosium (Dy)**, a heavy rare earth, is even more volatile. Prices have surged dramatically in response to Chinese export controls, with one report noting a nearly threefold increase in European prices in a single month during 2025. Prices in early 2026 were in the range of $263/kg to $930/kg, depending on the source and market. Average pricing for Dy2O3 was reported as $323/kg in 2023, down from a high of $410/kg in 2021.

Non-Chinese Supply Premiums: While there is significant strategic interest in establishing non-Chinese REE supply chains, quantified, consistent "premiums" for non-Chinese material are not well-documented in the provided sources. The value is currently expressed through government incentives, grants for new facilities, and strategic

offtake agreements rather than a clear spot market premium. The primary driver for sourcing non-Chinese REEs is supply chain security and risk mitigation, not a simple price arbitrage.

Executable Beachhead Model

Target Buyer and Pain Point:

- **Target Buyer**: Environmental, Health & Safety (EHS) Manager or Plant Manager at a new or expanding NdFeB magnet manufacturing facility in the US or EU.
- **Buyer's Pain**: The buyer is tasked with designing and operating a facility that complies with stringent environmental regulations (EU IED/BREF, US EPA) for wastewater and hazardous waste. Their primary pain is managing the significant **cost, complexity, and risk** associated with treating metal- and organic-laden wastewater and disposing of large volumes of hazardous sludge. They are motivated to find a solution that guarantees compliance, minimizes long-term liability, and reduces operational expenditure.

What They Pay For:

1. **Primary Value**: **Guaranteed environmental compliance**. This includes meeting effluent discharge limits for heavy metals and reducing or eliminating the generation of hazardous sludge, thereby avoiding high disposal costs and regulatory scrutiny. They are paying for a de-risked environmental management solution.
2. **Secondary Value**: **Resource recovery**. The value of the recovered REE oxides is a secondary benefit that improves the project's ROI. This can be structured as a revenue-sharing agreement or a discount on the service fee.

Purchase Triggers and Adoption Blockers:

- **Triggers**:
  - **New Plant Design Phase**: The most critical trigger is the opportunity to design the waste treatment system from the ground up ("design-in").
  - **Regulatory Enforcement**: An EPA or EU Member State enforcement action creates an immediate need.
  - **Rising Disposal Costs**: A significant increase in local hazardous waste landfill or transport costs.
- **Blockers**:
  - **Technology Risk**: Unproven performance on their specific, complex waste matrix. A paid pilot or on-site demonstration is essential to overcome this.
  - **High CAPEX**: A large upfront capital cost can be a barrier. A service-based model (e.g., pay-per-m^3 treated) or equipment lease can mitigate this.
  - **Process Integration Complexity**: Fear of disrupting core magnet manufacturing operations. The solution must have a simple, well-defined interface with the main plant.

Why Now: The timing is critical due to the **coincidence of two major trends**:

1. **Non-Chinese Supply Chain Buildout**: A wave of new NdFeB magnet plants is being built in the US and EU between 2024 and 2029. This provides a limited window to become the specified, default waste treatment solution for this new generation of facilities.
2. **Increasing Regulatory Pressure**: Regulators in the US and EU are tightening enforcement on industrial wastewater and are increasingly focused on a circular economy for critical materials, making compliance and resource recovery a higher priority for manufacturers.

Value Stack

The value of an integrated REE recovery solution is multi-layered, with the most immediate and bankable value derived from avoided compliance costs.

1. **Avoided Wastewater Treatment Cost**:
   - **Quantification**: OPEX for conventional metal finishing wastewater treatment ranges from **$1.03 to $4.88 per cubic meter**. For a plant generating 100 m^3/day, this translates to an avoided annual cost of **$37,500 - $178,120**. This value is realized if the proposed biological process replaces or simplifies the conventional chemical treatment train.
2. **Avoided Sludge Disposal / Reduced Hazardous Classification**:
   - **Quantification**: This is the largest component of the compliance value stack. Hazardous sludge disposal costs range from **$190 to over $310 per tonne** in the US and **€238+ per tonne** in the EU. If a plant generates 10 tons of wet sludge per month (a conservative estimate for a medium-sized unit), this amounts to an avoided annual cost of **$22,800 - $37,200+** in the US or **€28,560+** in the EU. A process that eliminates or de-classifies this sludge captures this value directly.
3. **Permitting/Pilot Pathway Value**:
   - **Quantification**: Difficult to quantify directly, but significant. By providing a "greener" solution that incorporates resource recovery, the technology can simplify and de-risk the environmental permitting process for new plants. This translates to **reduced project timelines** and **lower consulting fees**. RCRA permit acquisition can cost **$250,000 - $500,000** (1996 dollars); a solution that simplifies this process has immense value.
4. **Supply-Security Premium or Mandates**:
   - **Quantification**: Not directly quantifiable from market prices. The value is strategic. For manufacturers with government contracts (e.g., defense) or those seeking to market their products as having a "secure, non-Chinese supply chain," on-site REE recovery provides a powerful marketing and strategic advantage. The EU's Critical Raw Materials Act and US government initiatives signal a growing mandate for such capabilities.
5. **Recovered REE Commodity Value (Secondary)**:
   - **Quantification**: Highly variable. Assuming a 1,000 tonne/year magnet plant generates 20% waste (200 tonnes) with 25% REE content, this yields 50 tonnes of REE. At an average price of **$50-$100/kg** for an NdPr-rich oxide mix, this represents a potential revenue of **$2.5M - $5.0M per year**. This value is secondary and should be considered upside or used to offset service fees.

Data Gaps & Retrieval Plan

1. **Plant-Level Waste Volumes**:
   - **Missing Fact**: Actual generation volumes (m^3/day for liquids, tonnes/year for solids) and frequency of disposal for each waste stream from a commercial NdFeB magnet plant.
   - **Retrieval Plan**: Direct engagement with EHS managers at operational plants (e.g., Noveon Magnetics, MP Materials) under NDA. Request anonymized data from environmental permit applications or annual waste reports.
2. **Real Electroplating Effluent Chemistry**:
   - **Missing Fact**: Quantitative composition of REEs, competing metals, and specific organic additives (surfactants, brighteners, chelators) in electroplating effluent from an active magnet production line.

- **Retrieval Plan**: Propose a funded sampling and analysis program with a friendly magnet manufacturer. Obtain samples directly from their wastewater stream before pretreatment. Analyze via ICP-MS for metals and GC-MS/LC-MS for organics.

3. **Real Grinding Coolant Chemistry**:
   - **Missing Fact**: Same as above, but for grinding coolant/slurry. Specifically need concentrations of oils, surfactants, and suspended solids (wt%).
   - **Retrieval Plan**: Same as for electroplating effluent. Sample both the liquid coolant and the settled sludge/swarf for a complete characterization.
4. **Nd/Dy Effluent Limits in Local Permits**:
   - **Missing Fact**: Specific discharge limits for Nd, Pr, or Dy in any active NPDES (US) or IED (EU) permit for a rare earth processing or magnet facility.
   - **Retrieval Plan**: File Freedom of Information Act (FOIA) or EU equivalent requests for the full permits of MP Materials (Fort Worth, TX), USA Rare Earth (Stillwater, OK), and Neo Performance Materials (Narva, Estonia). Search the permit documents for any REE-specific limits.
5. **Cost of Sludge Disposal for Magnet Manufacturers**:
   - **Missing Fact**: Actual invoices or contract rates ($/tonne) for hazardous waste disposal from a US or EU magnet manufacturer.
   - **Retrieval Plan**: During industry outreach (see Gap 1), confidentially request anonymized cost data from EHS managers. Alternatively, contact hazardous waste disposal vendors (e.g., Veolia, Clean Harbors) and request budgetary quotes for disposal of F006 sludge with a representative composition.
6. **Wastewater Treatment OPEX Breakdown**:
   - **Missing Fact**: A detailed OPEX breakdown (chemicals, energy, labor, maintenance) for the wastewater treatment plant at an existing magnet facility.
   - **Retrieval Plan**: Direct engagement with plant managers under NDA. Request anonymized operational data. This is essential for a precise "avoided cost" calculation.
7. **Procurement Lead for Environmental Systems**:
   - **Missing Fact**: The specific job title and role responsible for specifying and purchasing wastewater/sludge technology at new-build magnet plants (e.g., is it the EPC firm, the corporate engineering team, or the local plant EHS lead?).
   - **Retrieval Plan**: Use LinkedIn to identify and interview engineering and EHS professionals at companies like Fluor, MP Materials, and VAC who were involved in recent plant buildouts.
8. **Permitting Pathway for Novel Treatment**:
   - **Missing Fact**: What specific pilot testing data and performance guarantees are required by a regulator (e.g., Texas TCEQ, Estonian Environment Agency) to approve a novel biological treatment system in a new plant's permit?
   - **Retrieval Plan**: Engage an environmental permitting consultant with experience in the target jurisdictions (Texas, South Carolina, Estonia). Conduct a "pre-application meeting" with the relevant regulatory agency to understand pilot requirements.
9. **Non-Chinese REE Oxide Premium**:

- **Missing Fact**: Quantified, transactional evidence of a price premium paid for Nd/Pr/Dy oxides produced outside of China.
- **Retrieval Plan**: Interview procurement specialists at major REE buyers (automotive OEMs, wind turbine manufacturers). Consult with commodity analysts at firms like Adamas Intelligence or Argus Media who track the physical market.

10. **Competing Ion Impact on Biology**:

    - **Missing Fact**: What are the specific tolerance limits of the proposed biological process to key contaminants identified in real waste streams (e.g., oils, surfactants, high concentrations of Fe, Cu)?
    - **Retrieval Plan**: Conduct internal, lab-scale inhibition studies. Spike synthetic REE media with varying concentrations of contaminants found in the literature (e.g., Syntilo 9902 coolant, specific surfactants) and measure the impact on REE recovery efficiency. This is a critical technical de-risking step.

1. What Can Be Bought or Built Today

Within the screened set, the most "buildable today" options for **manufacturing wastes** (swarf, grinding sludge/slurry, leachates arising from manufacturing scrap processing) are conventional **hydrometallurgical precipitation and solvent-extraction-style flowsheets**, plus **adsorbent polishing** of hydromet wastewaters; the evidence base in the excerpts is largely bench/lab, with one TEA/LCA-style industrial model for swarf processing. Acid-free dissolution with copper(II) nitrate is presented as an industrially modeled route for NdFeB **magnet swarf** (powders/filings from cutting and grinding), with downstream precipitation/filtration and oxide production steps (dissolution, filtration, ammonium hydroxide precipitation, oxalic acid precipitation, and roasting to mixed REOs).

For solvent extraction, a D2EHPA-in-kerosene route is demonstrated at bench scale with counter-current staging and quantified co-extraction behavior (Fe/B/Co), and a coupled approach using **solvent extraction plus extraction chromatography** is described to split REEs into separate fractions under sulfuric-acid elution conditions.

For "shop-floor" solids handling before hydromet (i.e., reducing iron interference before leaching), one recent process uses **flash reduction plus two-step wet magnetic separation** on **scraps produced during cutting, grinding, and polishing**, explicitly sourced from a magnet manufacturer (Ganzhou YouLi Technology). This is a pre-concentration route that increases REOs in a REO-rich phase and enriches an Fe-rich phase, with the stated benefit of minimizing Fe interference in subsequent leaching and avoiding substantial leaching waste residue rich in rare earth phases.

On the "removal from wastewater" end (rather than bulk recovery from solids), a nitrogen-rich hydrogel adsorbent (PEI-ALG@KLN) is presented for "waste NdFeB magnet hydrometallurgical wastewater," with reported Nd/Dy recovery and Fe rejection, suggesting a plausible **polishing/secondary recovery** unit operation after hydromet steps where an aqueous stream exists.

Named commercial entities appearing in the provided excerpts are primarily **suppliers of reagents or feedstock** rather than turnkey wastewater vendors: D2EHPA (97%) is stated as supplied by Sigma Aldrich, and manufacturer-provided scrap is attributed to Ganzhou YouLi Technology (Ganzhou, China).

2. Best Credible Benchmarks

The table below summarizes the best extractable, quantitative benchmarks in the provided excerpts, separated by whether the process targets bulk recovery from solids/leachates or adsorption from a wastewater stream. All values are reported as stated, with the evidence level indicated.

Technology class	Waste stream type	Best Nd metric	Best Dy metric	Best Pr metric	Evidence level
Hydromet (acid-free dissolution + precipitation to mixed REOs)	Magnet swarf from cutting/grinding	Overall REE extrac-tion/recovery efficiency 96%–99% (baseline 97%)	Dy treated as part of mixed REO output; scenarios assume 0.48% Dy in swarf (baseline) and vary Dy to 0% and 6% alongside 96%–99% leaching assumptions	Pr treated as part of mixed REO output; baseline assumes 3.33% Pr in swarf	TEA/LCA estimate based on lab data and equipment sizing
Hydromet + solvent extraction (D2EHPA SX)	PLS (leachate) from NdFeB waste	99.9% Nd extraction with three counter-current stages; >80% Nd stripped using 1.25 M H2SO4 at 25 ℃ (A/O 0.5)	>99.7% Dy extraction; >99% Dy extraction at pH 1.9	92% Pr extraction	Lab-scale (250 mL beaker SX/stripping)
Ion exchange/resin (extraction chromatography with D2EHPA-impregnated column)	PLS fed to column (Nd 462 mg/L; Pr 38 mg/L)	Nd+Pr fraction purity >98%; Nd fraction target below 0.01% Sm	Dy recovered last; Dy elution requires >0.9 M H2SO4	>50% Pr recovery in a third fraction with purity >75% (real PLS)	Lab-scale (50 µL injections; 1 mL/min eluate flowrate)
Hydromet (AMSS roasting + HCl leach + oxalate precipitation + calcination)	Swarf/slurry with coolant (AMSS includes 52.5% cooling reagent)	99% Nd dissolved in 0.5 N HCl; 99% Nd recovery; 18 g oxalic acid per L solution to precipitate 99% Nd; final product purity “over 99%”	Not found in provided excerpts (not reported quantitatively)	Not found in provided excerpts (not reported quantitatively)	Lab-scale (older, 2018)

Technology class	Waste stream type	Best Nd metric	Best Dy metric	Best Pr metric	Evidence level
Biological/biosorption (PEI-ALG@KLN hydrogel adsorbent)	"Waste NdFeB magnet hydrometallurgical wastewater"	Nd adsorption capacity 125 mg/g; Nd recovery rate >93%; capacity remains >85% after five adsorption–desorption cycles	Dy recovery rate >93% (reported jointly with Nd)	Not found in provided excerpts (not reported quantitatively)	Lab-scale
Other (flash reduction + magnetic separation)	Scraps from cutting/grinding/polishing	Not found in provided excerpts (element-specific Nd not reported)	Not found in provided excerpts	Not found in provided excerpts	Lab-scale (100 g processed; ~5 s flash reduction residence time)

Across the hydromet/SX pathways, the strongest **Nd and Dy extraction** numbers in the provided excerpts are from the staged D2EHPA solvent extraction experiments (99.9% Nd extraction; >99.7% Dy extraction). In contrast, the adsorption approach reports >93% recovery (Nd and Dy) and provides an Nd adsorption capacity (125 mg/g), but does not (in the provided excerpts) report wastewater discharge limits, residual concentrations, or full-stream mass balances that would be required to benchmark it against industrial effluent compliance needs.

For costs and energy, only the TEA/LCA-style swarf process provides explicit, extractable cost and energy assumptions in the provided excerpts: total cost of $1,092,585–$1,525,596/year (also expressed as ~$11 per kg of swarf processed or ~$29–$36 per kg of mixed REO produced), filtration energy assumed as 5.5 kWh/ton of dry material, and roasting energy reported as 16 kWh/kg of REO (4 kW muffle furnace; up to 4 hours per 1 kg REO).

3. Dominant Hidden Costs and Burdens

The dominant "hidden costs" visible in the excerpts are not line-item OPEX tables (which are generally absent from the lab-scale papers), but rather the **pretreatment requirements**, **interference management**, and **secondary solids/wastewater streams** implied by the chemistry and separations used.

A recurring burden is **iron management**. In the acid-free dissolution route, dissolution produces an REE-containing solution alongside copper and iron precipitates that must be separated by filtration, and an additional wastewater-treatment step is described to remove iron using calcium chloride and calcium hydroxide (prepared from calcium oxide). In the D2EHPA SX route, co-extraction is quantified (7% Fe, 4% B, 2% Co across three counter-current stages), and downstream handling is implied in the need for scrubbing (oxalic acid scrubbing enabling >86% Fe and >88% B scrubbing). In the same source family, the raffinate iron-removal step is described as potentially slow, with precipitation rates requiring control to maintain filterability, which signals a time-and-solids-handling penalty even when the solvent-extraction step itself performs well.

Another recurring burden is **coolant/organic content in manufacturing slurries**. The AMSS (acoustic magnet swarf slurry) is described as containing 52.5% cooling reagent, and oxidation roasting is described as serving to remove

cooling reagent and decrease iron dissolution, implying a significant thermal pretreatment and organic handling step to make leaching workable. Consistent with this, the AMSS route includes multiple high-temperature steps (e.g., oxidation roasting at 600 ℃ for 6 hours; calcination at 900 ℃ for 0.5 hour) and elevated-temperature leaching (e.g., 95 ℃ for 4 hours), which collectively imply substantial equipment and energy burdens even when direct Nd recovery is high in lab tests.

Pre-concentration by flash reduction plus magnetic separation is positioned as a way to **reduce Fe interference** and to avoid substantial leaching waste residue rich in rare earth phases, which is a different kind of "hidden cost" lever: shifting burden upstream into thermal reduction and physical separation to reduce downstream residue and leach complexity.

Finally, reagent circularity and loss can be a major burden. In the acid-free copper-nitrate dissolution TEA/LCA, copper nitrate is identified as a primary source of non-carcinogenics and ecotoxicity impacts, and copper nitrate recycling efficiency is stated as ~70%, implying ongoing reagent makeup and associated burden even under a "recycling" design.

4. Biological Win and Fail Conditions

The provided excerpts suggest two plausible incumbent benchmarks for a biological or bio-derived sorbent approach: (i) high extraction performance from hydromet/SX routes, and (ii) quantified Fe/REE discrimination. In staged D2EHPA solvent extraction, the benchmark extraction values reported are 99.9% Nd extraction and >99.7% Dy extraction under the stated counter-current staging, with quantified co-extraction (7% Fe, 4% B, 2% Co). In the PEI-ALG@KLN adsorption work, the benchmark values reported are >93% recovery for Nd and Dy, Nd capacity 125 mg/g, and Fe recovery only 5.04%, with >85% capacity retention after five adsorption–desorption cycles.

A credible set of **biological win conditions** (as hypotheses anchored to the extracted numbers) is therefore:

- The bio-based route should demonstrate Nd and Dy recovery that is at least comparable to the reported >93% recovery values, while maintaining strong Fe rejection comparable to Fe recovery of 5.04% in the adsorbent example.
- The bio-based route should show stability over multiple reuse cycles at least comparable to "capacity remains more than 85% after five adsorption–desorption cycles," because regeneration durability directly affects consumables and waste burdens for sorbent systems.
- If competing directly with solvent extraction for leachates, the bio-based route must be evaluated head-to-head against the very high extraction benchmarks reported for SX (99.9% Nd; >99.7% Dy), which likely necessitates either multi-stage contacting or a process design that can deliver comparable overall recovery at realistic flow and solids loads.

Corresponding **biological fail conditions** (again, hypotheses grounded in incumbent numbers) include:

- Strong performance collapse in the presence of iron-bearing matrices, because iron interference is a repeatedly surfaced operational issue across hydromet and SX routes (filtration of Fe/Cu precipitates, iron precipitation control for filterability, co-extraction that demands scrubbing).
- Inability to operate on coolant-laden or organic-contaminated streams without upstream destruction or removal, because manufacturing slurries can contain large coolant fractions (e.g., 52.5% cooling reagent in AMSS) and roasting is used explicitly to remove these organics before leaching in one manufacturing-slurry flowsheet.

5. Plant Baseline

The excerpts provide clear descriptions of **where manufacturing wastes arise** (e.g., magnet swarf as powders and filings from cutting and grinding; scraps produced during cutting, grinding, and polishing; grinding sludge from the sintered-magnet production process). They also show why these wastes are operationally challenging (e.g., AMSS described as 52.5% cooling reagent).

However, the provided excerpts do not contain an explicit, quantitative description of the prevailing in-plant "do-nothing" handling route (e.g., direct disposal routes, on-site neutralization-only, off-site hazardous waste classification, or internal returns to melt) for these manufacturing waste streams. A practical baseline-assessment approach consistent with the burdens described in the technical excerpts is to document: (i) where iron is removed (e.g., CaCl2 + Ca(OH)2 iron removal is described as a wastewater-treatment step in one TEA/LCA), (ii) whether coolant-rich slurries are thermally treated (oxidation roasting described as removing cooling reagent), and (iii) whether any plant already deploys physical pre-concentration to reduce leach residue burdens (flash reduction + magnetic separation described as avoiding substantial leaching waste residue).

6. Top Data Gaps

The gaps below are ranked by decision impact and time-to-resolve, and each includes a concrete retrieval plan. The list is constrained to what the excerpts explicitly do and do not report.

1) **Element-specific Dy and Pr performance on true manufacturing wastewaters (not just leachates)**. Dy and Pr metrics are often absent in the manufacturing slurry and sludge routes, while Nd is reported more frequently (e.g., AMSS Nd dissolution/recovery is quantified, but Dy/Pr are not). Retrieval plan: measure Nd/Pr/Dy concentrations (ICP-OES/ICP-MS) in influent/effluent streams for pilot trials on manufacturing coolants/rinse waters; look for plant wastewater monitoring reports and hydromet bleed-stream analyses.

2) **Residual concentrations and discharge metrics** for any technology applied to manufacturing wastewaters (mg/L after treatment). The adsorbent work reports recovery rates and capacity but not discharge endpoints in the excerpts. Retrieval plan: run batch and column tests reporting effluent concentration vs bed volumes; internal EHS compliance records typically capture discharge concentrations by outfall and permit parameter.

3) **Pretreatment requirements for adsorbent use**, including pH control, solids filtration, and regeneration chemistry. The adsorbent excerpts mention adsorption–desorption cycling but do not specify regeneration reagents. Retrieval plan: obtain full text methods and supporting information; request vendor-style regeneration protocol; verify via mass balance on eluate/reagent consumption per cycle.

4) **Full reagent consumption rates for hydromet leaching and precipitation steps**. One route reports oxalic acid demand as "18 grams" per "litre of solution," but broader reagent consumption (acid/base neutralization demand, scrubbing reagent) is not consistently enumerated in excerpts. Retrieval plan: from process design packages, compile reagent make-up and bleed; lab can measure titration curves (acid/base demand) and precipitation yields vs dosage.

5) **Secondary waste quantities and characterization** (filter cake mass, hazardous classification, Cu/Fe sludge). Filtration of Cu/Fe precipitates and iron precipitation control are described, but not the resulting sludge quantities. Retrieval plan: measure solids generation (kg dry solids per m^3 treated) and TCLP/metals; look for waste manifests and sludge dewatering logs from similar hydroxide precipitation systems.

6) **Economic breakdown of the TEA/LCA total cost into CAPEX and OPEX line items**. The TEA/LCA provides total annual cost and notes amortization over 20 years, but investment and operating line-item structure is not

extractable from the excerpts. Retrieval plan: access full TEA spreadsheet/supplement; request vendor EPC estimates for reactor/filtration/roaster; compare to the modeled reactor sizing example (198-gallon reactor for 100 tons/year).

7) **Solvent losses and solvent management requirements** for D2EHPA/kerosene systems. The SX excerpt specifies D2EHPA concentration and stripping conditions but not solvent loss rates. Retrieval plan: pilot-scale mixer-settler or column trial with organic inventory tracking; solvent loss typically appears in HSE solvent-emissions plans and OPEX reagent makeup tables.

8) **Performance under realistic chelators, oils, surfactants, and coolant additives** typical of grinding coolants and plating/finishing operations. The AMSS work demonstrates the magnitude of coolant fraction and uses roasting to remove it, but does not quantify organic interference in downstream separations. Retrieval plan: characterize TOC and specific additives in coolant and rinse waters; spike tests to quantify adsorption/SX performance impact at controlled compositions.

9) **Scale-up throughput for chromatography and adsorbent columns**. Chromatography is reported at 50 μL injections and 1 mL/min flow, which is far from plant scale. Retrieval plan: translate to column capacity by measuring dynamic binding capacity and breakthrough curves at relevant superficial velocities; vendors' column-resin datasheets and pilot skid specs typically provide these metrics.

10) **Selectivity coefficients (or separation factors) for Nd/Pr/Dy vs Fe/Co/B** across technologies. SX reports co-extraction and stripping behavior in %, while adsorption reports Fe recovery; neither provides a unified selectivity coefficient in the excerpts. Retrieval plan: compute separation factors from measured distribution coefficients in controlled experiments and report with co-ion composition (Fe, Co, B).

11) **Energy intensity for high-temperature pretreatments** in roasting-based manufacturing-waste flowsheets. Roasting temperatures and durations are given, and one TEA provides roasting energy intensity, but roasting energy is not consistently reported across routes. Retrieval plan: instrument roasters/furnaces for kWh per batch; energy reporting appears in TEA/LCA appendices and utility bills for thermal units.

12) **Impact of iron oxidation state control on downstream separations**. One leachate route adds 0.1 vol.% H2SO4 after filtration "to prevent the oxidation of ferrous iron," implying sensitivity to redox state. Retrieval plan: measure Fe(II)/Fe(III) speciation and its effect on extraction/adsorption; this is typically discussed in hydromet operating manuals and lab reports on PLS stability.

13) **Quantified benefit of physical pre-concentration on leaching residue and downstream reagent consumption**. Flash reduction + magnetic separation reports REOs enrichment and qualitative residue avoidance, but does not (in excerpts) quantify downstream reagent savings or residue mass. Retrieval plan: run paired leach tests (with/without pre-concentration) and quantify residue mass, leach kinetics, and acid consumption; capture in pilot campaign reports.

14) **Applicability to electroplating/finishing effluent and rinse waters**. None of the provided excerpts explicitly treats plating effluent or rinse waters. Retrieval plan: search and screen plating-effluent rare-earth removal studies; in-plant validation via composite sampling at plating rinse lines and bench adsorption/precipitation trials with co-ions (Ni, Cu, Zn) as present.

15) **Commercial vendor offerings for manufacturing-waste recovery** (as opposed to EOL recycling plants). The excerpts include reagent suppliers and a manufacturer scrap source, but do not include turnkey vendor process datasheets for coolant/sludge/wastewater treatment. Retrieval plan: collect vendor technical datasheets and case studies; the fastest path is structured vendor calls requesting: treated flow (m^3/day), inlet metals (mg/L),

outlet metals (mg/L), resin/sorbent life, consumables, and installed cost basis; then validate via a short on-site trial using plant composite samples.

Synthesis

The strongest quantitative benchmarks in the provided excerpts for manufacturing-waste-derived streams come from a bench **D2EHPA solvent extraction** route (99.9% Nd extraction; >99.7% Dy extraction) and a TEA/LCA-modeled **acid-free dissolution** route (overall REE extraction/recovery 96%–99%, baseline 97%). For wastewater-side removal, a bio-derived hydrogel adsorbent shows >93% recovery of Nd and Dy and reports Nd capacity of 125 mg/g with Fe recovery of 5.04%, suggesting promise for selective polishing but with missing discharge-style metrics in the excerpts. Across routes, the dominant hidden burdens are iron handling (filtration/precipitation and slow iron-removal kinetics), coolant/organic content (AMSS at 52.5% cooling reagent driving roasting pretreatment), and reagent circularity/loss (copper nitrate recycling efficiency ~70% and identified as an ecotoxicity driver). A biological approach plausibly "wins" if it maintains >93% Nd/Dy recovery with strong Fe rejection and multi-cycle stability (>85% capacity after five cycles), and "fails" if it cannot tolerate iron-rich matrices or coolant-laden streams without energy-intensive pretreatment comparable to oxidation roasting. The excerpts clearly identify where wastes originate (cutting/grinding swarf; grinding sludge; polishing scraps) but do not provide a direct, quantified description of the prevailing in-plant "do-nothing" baseline, so establishing a baseline should begin by mapping existing iron removal, coolant handling, and residue management practices against the unit operations explicitly described in the recovery flowsheets (e.g., CaCl2 + Ca(OH)2 iron removal; oxidation roasting for coolant removal; pre-concentration to avoid leach residues).

Plant Capacities

This section details announced and operational NdFeB magnet manufacturing capacity, focusing on new buildouts in the US, EU, and Vietnam.

Metric	Value	Unit	Year	Geography	Waste Stream	Conditions	Evidence Level
Production Capacity (MP Materials)	1,000	tonnes/year	2025	Fort Worth, TX, USA	Not specified	Initial capacity at "Independence" facility; gradual ramp beginning late 2025	Commercial
Production Capacity (MP Materials, 10X Facility)	10,000	metric tons/year	2028 (expected)	Mountain Pass, CA, USA	Not specified	Planned second domestic magnet facility, "10X Facility"	Policy/market estimate
Production Capacity (E-VAC Magnetics)	~1,500	tonnes/year	2025 (late fall)	Sumter, SC, USA	Not specified	Sintered NdFeB magnet plant under construction	Commercial
Production Capacity (USA Rare Earth)	1,200 (initial)	tonnes/year	2025-2026	Stillwater, OK, USA	Not specified	Ramping to 4,800 tonnes/year by ~2026	Commercial
Production Capacity (Noveon Magnetics)	>2,000	tons/year	2020 (ramped up)	San Marcos, TX, USA	Not specified	Operational facility; expanding capacity beyond 2,000 tons/year	Commercial
Production Capacity (Neo Performance Materials)	2,000	tonnes/year	2025	Narva, Estonia	Not specified	Phase 1 capacity; plans to expand to 5,000 tonnes/year	Commercial

Metric	Value	Unit	Year	Geography	Waste Stream	Conditions	Evidence Level
Production Capacity (HyProMag USA)	1,552 (total)	metric tons Nd-FeB/year	2027-2029	Dallas-Fort Worth, TX, USA	Not specified	Texas Hub; comprises 941 tpa recycled magnets and 611 tpa co-products	Policy/market estimate
Production Capacity (POSCO International JV)	3,000 (initial)	tons/year	Not specified	USA (TX, TN, or AZ)	Not specified	Planned factory with future scale-up to 5,000 tons/year	Policy/market estimate
Production Capacity (SGI Vina)	4,000-5,000	tons/year	2025	Quang Nam Province, Vietnam	Not specified	Stage I (1,600 tpa) in Q1-24; Stage II (4,000 tpa) by mid-2025	Commercial
Production Capacity (Shin-Etsu Vietnam)	2,200	tons/year	2018	Hai Phong Province, Vietnam	Not specified	Expansion completed in mid-2018, doubling capacity from 1,100 tpa	Commercial
Production Capacity (Daido Steel)	1,800	tons/year	2030	Gifu Prefecture, Japan	Not specified	Expansion to 150 tons/month (1,800 tpa) planned for fiscal 2030	Policy/market estimate

Waste Chemistry

This section outlines the chemical composition of various NdFeB manufacturing waste streams as reported in academic and technical literature.

Metric	Value	Unit	Year	Geography	Waste Stream	Conditions	Evidence Level
Total REE Concentration	up to 30	wt%	2024	Not specified	Machining waste (cutting/grinding)	Sintered Nd-Pr-Fe-B magnet residue	Commercial
Nd Concentration	27.19	wt%	2021	USA	Magnet Swarf	Baseline composition assumed for TEA/LCA model	Lab-scale
Pr Concentration	3.33	wt%	2021	USA	Magnet Swarf	Baseline composition assumed for TEA/LCA model	Lab-scale
Dy Concentration	0.48	wt%	2021	USA	Magnet Swarf	Baseline composition assumed for TEA/LCA model	Lab-scale
Fe Concentration	66.4	wt%	2023	Not specified	Grinding residue	Industrial machine grinding of N38 magnet	Commercial
Fe Concentration	52.4	wt%	2023	Not specified	Diamond cut residue	Laboratory cutting bench sample	Lab-scale
Fe Concentration	67.3135	wt%	2016	China	NdFeB Sludge	Pretreatment powder from industrial sludge (XRF analysis)	Commercial

Metric	Value	Unit	Year	Geography	Waste Stream	Conditions	Evidence Level
Fe Concentration	60.3	wt%	2018	Not specified	Sintering waste	Ground manufacturing waste from sintering stage	Commercial
Co Concentration	1.1343	wt%	2016	China	NdFeB Sludge	Pretreatment powder from industrial sludge (XRF analysis)	Commercial
Organics Content	52.5	wt%	2014	Not specified	Swarf Slurry (AMSS)	Stated as ”cooling reagent” in the slurry composition	Commercial
Nd Concentration (leachate)	9.01	g/L	2019	Not specified	NdFeB Scrap Leachate	Scrap dissolved in nitric acid before hydrothermal treatment	Lab-scale
Fe Concentration (leachate)	10.53	g/L	2019	Not specified	NdFeB Scrap Leachate	Scrap dissolved in nitric acid before hydrothermal treatment	Lab-scale
Nd Concentration (leachate)	1.63	g/L	2020	India	EoL HDD Magnet Leachate	Leachate from HCl extraction of ground HDD magnet powder	Lab-scale

Metric	Value	Unit	Year	Geography	Waste Stream	Conditions	Evidence Level
Fe Concentration (leachate)	~4.5	g/L	2020	India	EoL HDD Magnet Leachate	Calculated from reported 45.76 wt% Fe in solids at 10 g/L pulp density	Lab-scale

Compliance Economics

This section provides cost data for industrial wastewater treatment and hazardous waste disposal, which serve as a baseline for the compliance-driven business case.

Metric	Value	Unit	Year	Geography	Waste stream	Conditions	Evidence Level
Hazardous Sludge Disposal	15,000-25,000	INR/tonne	2025	India	ETP Sludge	Disposal at CPCB-approved TSDFs	Commercial
Non-Hazardous Sludge Disposal	3,000-8,000	INR/tonne	2025	India	ETP Sludge	Not specified	Commercial
Hazardous Waste Disposal	310	USD/ton	2021	California, USA	Restricted hazardous waste	Average disposal fee	Commercial
Hazardous Waste Incineration	529	USD/ton	1998	USA	General hazardous waste	Average cost from USACE survey of RCRA-permitted facilities	Commercial
Wastewater Treatment OPEX	3.09	USD/m³	2020	Turkey	Metalworking wastewater	Bench-scale continuous electrocoagulation process	Lab-scale

Metric	Value	Unit	Year	Geography	Waste stream	Conditions	Evidence Level
Hazardous Liquid Disposal	1.84-3.11	USD/gallon	1998	USA	Hazardous wastewater	Cost for discharge to POTW/TSD, adjusted to 2000 USD	Policy/market estimate
Permit Acquisition Cost	250,000-500,000	USD	1996	USA	Hazardous waste incinerator	Estimated cost to obtain a RCRA or BIF permit	Policy/market estimate
Compliance Testing Cost	~300,000	USD	1996	USA	Hazardous waste incinerator	Cost per trial burn required for permit, based on vendor quotes	Commercial
Annual Compli-ance Overhead	50,000-100,000	INR/year	2025	India	ETP Sludge	Documentation & compliance for medium-sized industrial units	Commercial

REE Pricing

This section summarizes recent and historical price data for key rare earth elements, highlighting volatility. This value is considered secondary upside.

Metric	Value	Unit	Year	Geography	Waste Stream	Conditions	Evidence Level
Nd Oxide Price	49,763	USD/metric ton	2020	Global	Not specified	Historical average price	Policy/market estimate
Nd Oxide Price (Forecast)	77,500	USD/metric ton	2025	Global	Not specified	Statista forecast	Policy/market estimate

Metric	Value	Unit	Year	Geography	Waste Stream	Conditions	Evidence Level
NdPr Oxide Price (Forecast)	90,000	USD/tonne	2026	Global	Not specified	BMI average yearly price forecast	Policy/market estimate
Dy Price	930.70	USD/kg	2026	Not specified	Not specified	Retail price for private investors, last updated Feb 06, 2026	Market Estimate
Dy Oxide Price	263.34	USD/kg	2026	Northeast Asia	Not specified	Market price for January 2026	Market Estimate
Dy Price Volatility	+105.05	%	2026	Not specified	Not specified	Year-to-date price change as of Feb 06, 2026	Market Estimate
Nd Price	194.40	USD/kg	2026	Not specified	Not specified	Retail price for private investors, last updated Feb 06, 2026	Market Estimate
Nd Price Volatility	+199.08	%	2020-2026	Not specified	Not specified	Price change since January 2020	Market Estimate
NdPr Price (Historical)	360,500	CNY/Tonne	2020	China	Not specified	Historical spot price on 2020-01-02	Market Estimate
NdPr Price (Historical Peak)	1,520,000	CNY/Tonne	2022	China	Not specified	All-time high price in February 2022	Market Estimate